

Stochastic and robust truck-and-drone routing problems with deadlines: A Benders decomposition approach[☆]

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ABSTRACT

Time-definite delivery service requires efficient and punctual delivery within specified deadlines. Enhancing the service level of delivery in last-mile logistics prompted the application of a truck-and-drone cooperative system. However, the inherent uncertainty in travel time contributes to frequent service delays. To address this challenge, our study addresses the *stochastic and robust truck-and-drone routing problems* with deadlines, aiming to minimize service delay risk under uncertainty. We leverage the sample average approximation (SAA) and robust optimization (RO) approaches to handle uncertainty in scenarios involving both big and small datasets. Benders decomposition (BD) algorithms are developed to solve the SAA and RO truck-and-drone routing problems efficiently. Numerical experiments showcase the algorithms' effectiveness and reveal interesting insights. These findings suggest the importance of tailoring the approach to data availability for optimizing truck-and-drone delivery under uncertainty. Furthermore, the proposed model formulations and algorithms are generalized to address truck-and-drone routing problems under more complex scenarios, such as that with uncertainty of service time, or with nonlinear objective functions.

1. Introduction

The rapid growth of e-commerce, coupled with rising consumer demand for efficient and timely deliveries, presents both opportunities and challenges in last-mile logistics (Lemardelé et al., 2021). In order to gain an advantage in fierce competition, companies start offering delivery services with increasingly narrow deadlines (Ulmer, 2017). This requires companies to operate their warehouses and delivery networks economically and efficiently to deliver packages within the promised deadline. The intricate logistics involved in last-mile delivery, encompassing complex transportation and personnel scheduling issues, has perennially posed challenges for the industry. To address these challenges, drones have been introduced into last-mile delivery systems, offering the

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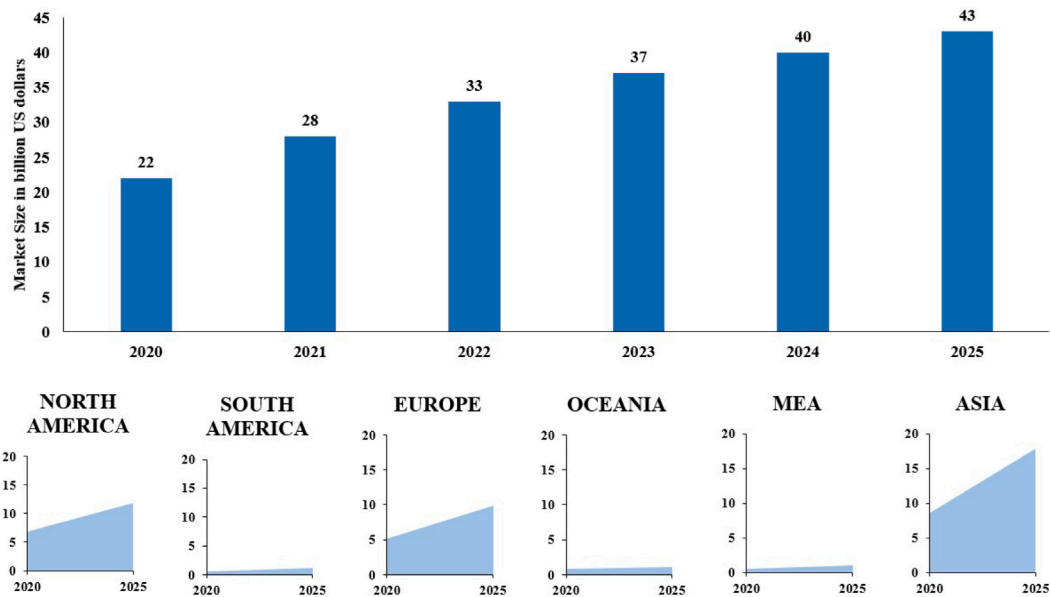


Fig. 1. Global drone market size and forecast from 2020 to 2025 (in billion US dollars).

advantages of bypassing traffic congestion and delivering items more rapidly through direct routes. According to a recent report,¹ global drone market size grows rapidly in recent years, from 22 billion \$ in 2020 to 37 billion \$ in 2023, as shown in Fig. 1. It is expected that the market size will reach 43 billion \$ in 2025. The small pictures show the drone market size data of all the main continents. One can see that, the drone market is rapidly growing in Asia, Europe, and North America. In USA, Amazon launched the Prime Air delivery system that aim to provide customers drone-delivery service within 30 min (Amazon, 2019; Liu et al., 2021). Nevertheless, current technological limitations, such as cargo capacity and endurance, restrict the complete substitution of traditional delivery methods with drone delivery. As a pragmatic solution, the truck-and-drone cooperative delivery system emerges as an efficient alternative that is poised to reshape the logistics industry in the foreseeable future (Gao et al., 2023).

The cooperative delivery of truck-and-drone has been applied in practice by multiple courier companies such as FedEx, Amazon, and Meituan (Cary and Bose, 2016). In complex and ever-changing market and emergency rescue scenarios, truck-and-drone delivery systems can adopt flexible and diverse hybrid models. A typical hybrid delivery model is performed as follows, the truck stops to serve a customer, meanwhile, it releases the drone with a package to serve another customer, and then recovers the drone at the same location (Poikonen et al., 2019). In this study, we will also focus on this hybrid truck-and-drone delivery model. This model can be extended to cooperative delivery models with a single truck and multiple drones (Murray and Chu, 2015; Poikonen and Golden, 2020), or multiple trucks and multiple drones (Sacramento et al., 2019).

As contemporary consumers increasingly demand real-time and convenient purchasing experiences, the concept of time-definite delivery service has gained prominence in the retail industry. This service entails providing deliveries within specific deadlines, necessitating robust logistics integration and a stellar industry reputation. Within the realm of time-definite delivery services, the focus is now on enhancing the delivery service level (Alumur et al., 2012). However, the practicality of truck-and-drone delivery is challenged by various uncertainties, such as the uncertainty of weather conditions, the uncertainty of travel time, and so on. In real-world scenario, the uncertainties often lead to service delays. The resultant delays have garnered significant attention from logistics enterprises, yet there is currently a paucity of literature addressing truck-and-drone delivery problems from the perspective of reducing service delays and improving service levels.

To bridge this gap, we propose a truck-and-drone routing problem with the decision objective of finding feasible routing solutions that minimize service delay risk. Traditionally, stochastic programming (SO) approaches model travel time uncertainty using probability distribution functions, which can be challenging to accurately capture in practice. Another common approach, robust optimization (RO), offers resilience with limited information. In this study, we take a data-driven approach to address travel time uncertainty. When dealing with large datasets, we can estimate the true travel time distribution by leveraging samples. Conversely, for smaller datasets, we construct an uncertain set of travel time by using samples to estimate key parameters.

In summary, this study makes the following distinct contributions:

- **Novel Decision Objective:** We are the first to formulate the truck-and-drone routing problem with deadlines (TDRP-D) with the service delay risk index as the objective. This metric effectively measures the severity of service lateness under uncertain environments, aligning with the decision-makers' focus on service level improvement when providing time-definite deliveries.

¹ <https://www.xflyfire.com/en/h-nd-198.html>.

- **Data-driven Uncertainty Handling:** Our data-driven approach tackles travel time uncertainty. For large datasets, the true distribution is approximated using a sample-based empirical distribution, leading to a approach to approximate the stochastic TDRP-D model (STDRP-D). For smaller datasets, we estimate parameters for a budget uncertain set, formulating the robust TDRP-D model (RTDRP-D) for decision-making under uncertainty. More importantly, the data-driven approach makes the models and algorithms proposed in this paper applicable to truck-and-drone routing problems with more complex nonlinear objective functions.
- **Benders Decomposition Algorithms:** We develop Benders decomposition algorithms to solve the STDRP-D and RTDRP-D models, respectively. Computational results demonstrate that the RTDRP-D excels with small datasets, while the STDRP-D outperforms all benchmarks with large datasets. Additionally, our proposed Benders decomposition algorithms exhibit efficient computational performance.
- **Valid Inequalities:** We explore several valid inequalities to boost the computational procedure of truck-and-drone routing problems. Per our knowledge, most of the valid inequalities are the first to appear in literature on truck-and-drone routing problems. We demonstrate their efficiency through numerical experiments.

The remainder of this paper is organized as follows. Section 2 reviews literature closely related to this study. Section 3 describes the TDRP-D problem, and Section 4 introduces the truck-and-drone multi-commodity flow formulation and defines the feasible set of routing solutions. In Section 5, we formulate the STDRP-D and RTDRP-D models. Section 6 develops Benders decomposition algorithms for solving the STDRP-D and RTDRP-D models, respectively. In Section 7, we extend the model to handle problems in several complex scenarios. Section 8 reports the numerical results. Finally, we conclude this article in Section 9.

2. Literature review

In this section, we review the literature related to this study. First, we summarize existing research on truck-and-drone routing problems. Next, we review the application of robust and data-driven optimization in routing problems. Finally, we compare this study with several related works published in recent years.

2.1. Truck-and-drone routing problems

Truck-and-drone routing problems (TDRP), as an extension of classic vehicle routing problems (VRP), have garnered widespread attention from the academic community in recent years. Both TDRP and VRP focus on delivering packages to customers with minimal transportation costs (or time) by choosing an optimal routing solution. Differently, TDRP not only focuses on routing decision-making, but also involves effectively coordinating the delivery of trucks and drones. The exploration of this topic can be traced back to [Murray and Chu \(2015\)](#), where the TDRP with a single truck and a single drone was initially studied. Since then, various TDRP variants have been proposed and discussed in the literature. [Karak and Abdelghany \(2019\)](#) explores a hybrid TDRP with pick-up and delivery services, while [Kitjacharoenchai et al. \(2020\)](#) introduces a synchronized truck-and-multiple-drones routing problem where a single truck launches multiple drones to serve customers and retrieves them after completing the service. [Salama and Srinivas \(2022\)](#) studies the TDRP with a truck in tandem with multiple drones, allowing the truck to stop at non-customer locations to launch and retrieve drones. [Roberti and Ruthmair \(2021\)](#) considers the TDRP with a single truck teamed up with a single drone providing delivery service to multiple customers. [Shen et al. \(2021\)](#) investigates the TDRP for a delivery system with multiple warehouses. [Zhen et al. \(2023\)](#) investigates the TDRP for a cooperative delivery system with multiple trucks and multiple drones. More recently, [Meng et al. \(2023\)](#) proposes a new variant of the combined TDRP in which multiple trucks, each equipped with a drone, provide pickup and delivery services for customers. Besides, various research efforts have imposed time window constraints on routing problems ([Desaulniers et al., 2008](#); [Alumur et al., 2012](#); [Baldacci et al., 2012](#); [Paradiso et al., 2020](#); [Yang et al., 2021](#); [Zhang et al., 2021](#)). However, only a limited number of studies, including [Yin et al. \(2023\)](#), have discussed the TDRP with time windows (TDRP-D) so far.

Several existing TDRP models predominantly focus on minimizing total travel time ([Kitjacharoenchai et al., 2020](#); [Cavani et al., 2021](#); [Roberti and Ruthmair, 2021](#); [Gómez-Lagos et al., 2022](#); [Salama and Srinivas, 2022](#); [Zhao et al., 2022](#)). [Moshref-Javadi et al. \(2020\)](#) aims at minimizing the total customer waiting time by determining the routes of the truck and the drones. Another stream of models on TDRP strives to minimize routing or other associated costs ([Alumur et al., 2012](#); [Karak and Abdelghany, 2019](#); [Zhen et al., 2023](#); [Tamke and Buscher, 2023](#)). For instance, [Yang et al. \(2021\)](#) proposes a model to minimize total routing and tardiness costs, while both [Dukkanci et al. \(2021\)](#) and [Gao et al. \(2023\)](#) aim at minimizing the total operational costs brought about by energy consumption of the truck-and-drone, among other factors. Some models even attempt to achieve dual objectives ([Liu and Sun, 2022](#)).

TDRP models in the literature are commonly formulated as Mixed Integer Linear Programming (MILP) problems ([Karak and Abdelghany, 2019](#); [Kitjacharoenchai et al., 2020](#); [Moshref-Javadi et al., 2020](#); [Cavani et al., 2021](#); [Roberti and Ruthmair, 2021](#); [Tamke and Buscher, 2021](#); [Gómez-Lagos et al., 2022](#); [Meng et al., 2023](#); [Tamke and Buscher, 2023](#); [Zhen et al., 2023](#)). However, solving TDRP models for large problem sizes can be time-consuming. To address this issue, several exact solution procedures have been proposed in the literature, such as branch-and-bound (BnB) ([Poikonen et al., 2019](#)), branch-and-cut (BnC) ([Cavani et al., 2021](#); [Tamke and Buscher, 2021](#)), and branch-price-and-cut (BnPC) ([Li et al., 2010](#); [Roberti and Ruthmair, 2021](#); [Rostami et al., 2021](#); [Zhen et al., 2023](#)).

Table 1

Overview of related works in recent years.

Paper	Delivery type	TW/D ^a	Objective ^b	DD ^c	Uncertainty	Models ^d	Methodologies ^e
Poikonen et al. (2019)	Truck-and-drone	–	Time	–	–	DBF	BnB
Zhang et al. (2019)	Truck	✓	Risk	✓	✓	SAA,DRO	BD
Cavani et al. (2021)	Drone	–	Time	–	–	MILP	BnC
Dukkanci et al. (2021)	Truck-and-drone	✓	Cost	–	–	SOC	Solve Directly
Roberti and Ruthmair (2021)	Truck-and-drone	–	Time	–	–	MILP	DP, BnP
Tamke and Buscher (2021)	Truck-and-drone	–	Time	–	–	MILP	BnC
Yang et al. (2021)	Truck	✓	Cost	–	–	MILP	BnPC
Zhang et al. (2021)	Truck	✓	Risk	✓	✓	DRO	BnC, Heuristics
Liu and Sun (2022)	Drone	–	Cost	–	–	BCP	BE
Zhao et al. (2022)	Truck-and-drone	–	Time	–	✓	RO,SOC	Heuristics
Yin et al. (2023)	Truck-and-drone	✓	Cost	–	–	MILP	BnPC
This work	Truck-and-drone	✓	Risk	✓	✓	SAA, RO	BD

^a **TW/D**: Time windows or deadlines.^b **Time**: Minimizing the traveling (waiting or other) time; **Cost**: Minimizing the traveling or waiting (or other) costs; **Risk**: Service delay risk with respect to the time windows.^c **DD**: Data-driven.^d **DRO**: Distributionally robust optimization; **DP**: Dynamic Programming; **MILP**: Mixed integer linear programming; **DBF**: Dynamics of Bellman-Ford Equations; **BCP**: Brownian control problem.^e **BD**: Benders decomposition; **BnB**: Branch-and-Bound; **BnC**: Branch-and-Cut; **BnP**: Branch-and-price; **BnPC**: Branch-and-price-and-cut; **BE**: Bellman equation approach.

Moreover, many other efficient heuristic algorithms have been developed for truck-and-drone routing problems in different scenarios. Kitjacharoenchai et al. (2020) designed a large neighborhood search (LNS) algorithm to solve the two echelon vehicle routing problem with drones. Numerical results demonstrated that the heuristic algorithm outperforms the MIP method in both small (with 8 customers) and medium-size (about 40 customers) instances. Karak and Abdelghany (2019) extended the classic Clarke-Wright algorithm to solve the hybrid vehicle-drone routing problem for pick-up and delivery services. The average time it takes for this algorithm to solve problem instances with 50 customers is 21.728 s, greatly less than the time spent by the benchmarks. There are also heuristics developed for routing problem with multiple trucks and drones, such as the truck-and-drone routing algorithm (TDRA) (Moshref-Javadi et al., 2020), and the optimization-enabled search algorithm (Salama and Srinivas, 2022). The previous algorithm can solve the truck and drone routing problem with up to 20 customers in approximately 62 s, while the latter algorithm can solve the problem with 25 customers in approximately 501 s. Though the above heuristic algorithms has been validated to have high computational efficiency compared to benchmarks, their limitations cannot be ignored. First, they all focus on deterministic truck-and-drone routing problems, and the practicality of these algorithms still needs to be verified in uncertain environments. In addition, these algorithms are only used in the literature to handle routing problems with linear objective functions, and their performance in nonlinear routing problems has not yet been demonstrated.

2.2. Stochastic and robust optimization

The uncertainty of travel time is a significant factor in the delivery process of truck-and-drones. VRP models incorporating uncertainty are typically formulated in either stochastic or robust optimization frameworks. Stochastic VRP describes uncertainty using a known probability distribution and aims to minimize the expected routing cost or makespan. However, solving large-scale stochastic VRP problems is generally challenging. To address this issue, a commonly used approach involves generating a finite set of samples from the distribution and then solving the routing problem over these samples as an approximation to the stochastic problem (Adulyasak and Jaillet, 2016; Zhang et al., 2019, 2021).

Robust VRP describes uncertainty using a specified uncertain set and seeks the optimal routing that minimizes the worst-case cost or makespan. Lu and Gzara (2019) adopts a cardinality-constrained uncertain set to formulate the robust VRP with time windows (RVRP-TW) and proposes an effective Branch-and-Price-and-Cut (BnPC) algorithm to solve the model. Another approach, proposed by Zhang et al. (2021), utilizes a Wasserstein ambiguity set to describe the uncertainty of the travel time distribution. They introduce a distributionally robust VRP-TW model aimed at minimizing the service fulfillment risk index (SRI). The authors develop an exact Branch-and-Cut (BnC) method and a heuristic algorithm to expedite the solving procedure for the model.

The RO approach has also been employed to handle uncertainty in TDRP problems. Zhao et al. (2022) investigates a robust traveling salesman problem with multiple drones (RTSP-mD), formulated as a second-order conic programming (SOC) to minimize makespan and synchronization risk simultaneously. On the other hand, Yin et al. (2023) explores the RTDRP considering uncertainty in both travel time and demand. The model is formulated based on the widely used budget uncertain set and is solved by an enhanced BnPC algorithm.

2.3. An overview of closely related literature

In this subsection, we provide a summary of the literature closely related to this work. An overview of related works is presented in Table 1, comparing literature in terms of seven key aspects.

In the realm of the underlying problem, [Roberti and Ruthmair \(2021\)](#) emerges as the closest literature to this study. Both works discuss a Truck-and-Drone Routing Problem (TDRP) where a single truck and a single drone collaborate to serve multiple customers. The feasible routing solution in this study is formulated by referring to the constraints defined in [Roberti and Ruthmair \(2021\)](#). However, several distinctions set this study apart. *First*, our focus is on the TDRP from a time-definite service perspective, emphasizing the minimization of service delay risk, whereas Roberti's work aims to minimize makespan. *Second*, we incorporate the uncertainty of travel time in this study, while [Roberti and Ruthmair \(2021\)](#) operates under a deterministic scenario. Consequently, we formulate the problem using SAA and RO approaches, while [Roberti and Ruthmair \(2021\)](#) focuses on a deterministic MILP problem. *Finally*, we develop effective BD algorithms to solve the problem, while [Roberti and Ruthmair \(2021\)](#) employs a BnP method.

This paper essentially differs from the existing literature in the following aspects. First, while all TDRP models in the literature concentrate on minimizing total operation costs or travel (or other) times, this study delve into routing decisions that minimize service delay risk. Regarding methodology, [Zhang et al. \(2019, 2021\)](#) serve as the most inspiring literature for this work. The objectives of these works are to minimize service delay risk concerning time windows. Furthermore, all these works formulate problems through the RO or DRO approach. In this study, we generalize the methodologies proposed in [Zhang et al. \(2019, 2021\)](#) to solve the TDRP with deadlines.

Second, this study stands out from existing literature on TDRP in that we consider the uncertainty of drone travel time, whereas most previous studies assumed that drone travel time is deterministic. The process of drone delivery faces many uncertainties, such as wind conditions, which affect the travel time of drones to their destinations ([Cheng et al., 2024](#)). Therefore, incorporating the uncertainty of travel time into the study of drone routing problems is of practical value. Moreover, different from existing literature, this paper delve into truck-and-drone routing problem from a service level perspective, that is, the delivery service should be provided before the deadline. Under this setting, the uncertainty of drone flight time is not negligible theoretically, as even a few seconds of delay may lead to service violation. Therefore, it is worth considering the uncertainty of drone flight time in this study.

Finally, different from existing literature, this paper employs a data-driven approach to characterize the uncertainty of drone travel time. When the size of sample data is large, we formulate a sample-based empirical distribution to approximate the true distribution of drone travel time and solve the stochastic truck-and-drone routing problem. In contrast, when the sample size is small, we use a budget uncertain set with parameters estimated from the data to formulate a robust optimization model for truck-and-drone routing problems. This approach provides us more flexibility in handling the uncertainty of drone travel time. Moreover, the data-driven approach also enables us to extend the model and the solving procedure to scenarios with nonlinear objective functions.

3. Problem description of the TDRP-D

The truck and drone routing problem with deadlines (TDRP-D) is formally defined on a complete directed graph $G = (\mathcal{V}, \mathcal{A})$. The set of nodes is defined as $\mathcal{V} = \{0, 0'\} \cup \mathcal{N}$. Nodes 0 and $0'$ represent the same depot, where a single truck equipped with a single drone initiates and concludes the travel, respectively. The set of nodes in $\mathcal{N} = \{1, 2, \dots, N\}$ comprises the N customers, all of which can be serviced by the truck or the drone. The set of arcs is defined as $\mathcal{A} = \{(0, i) \mid i \in \mathcal{N}\} \cup \{(i, j) \mid i \neq j \in \mathcal{N}\} \cup \{(i, 0') \mid i \in \mathcal{N}\}$.

In real-world logistics operations, many companies offer customers time-definite delivery services, requiring delivery within a specified deadline. Each node $i \in \mathcal{N}$ is associated with a deadline $\bar{\tau}_i \in \mathbb{R}_+ \cup \{+\infty\}$, which represents the latest time for providing delivery service to customer i . It is possible that some customers do not specify deadlines. In such cases, we $\bar{\tau}_i = +\infty$. Denote by $\bar{\mathcal{N}} := \{i \in \mathcal{N} \mid \bar{\tau}_i < +\infty\}$, the set of customers with positive finite deadlines. For an arc $(i, j) \in \mathcal{A}$, let z_{ij}^T and z_{ij}^D be the travel time of the truck and the drone, respectively. For each $i \in \mathcal{N}$, denote by s_i^T, s_i^D the service time of node i provided by the truck and the drone, respectively. To simplify the analysis, we assume that the service time of each node is deterministic, as the uncertainty of service time can be well incorporated into the upcoming models and analysis, which will be discussed in Section 7.1.

The illustration in [Fig. 2](#) exemplifies a TDRP-D scenario involving nine customers and a single depot. The truck and the drone depart from the depot (node 0) at time $t_0 = 0$, corresponding to the real-world time 8:00. The truck travels from node 0 to serve customer 1, whose service deadline is $\bar{\tau}_1 = 120$, while simultaneously, the drone is launched from node 0 to serve customer 3 with the same deadline. After completing their services, both the truck and the drone travel from node 1 and node 3, respectively, to converge at node 2. The arrival order at node 2 determines whether the truck or the drone needs to wait for the other to rejoin.

The departure time from node 2 is determined by the maximum of the following two time values: the time required for the truck to travel the route $0 \rightarrow 1 \rightarrow 2$ plus the truck's service time for customer 1, i.e., $s_1^T + z_{01}^T + z_{12}^T$; the time required for the drone to travel the route $0 \rightarrow 3 \rightarrow 2$ plus the drone's service time for customer 3, i.e., $s_3^D + z_{03}^D + z_{32}^D$. Once synchronized, the truck and drone leave together from node 2 to serve customer 4.

The subsequent steps involve the truck launching the drone at node 4 to serve customer 5, followed by waiting for the drone to return. Afterward, they move together to serve customer 6. Upon completing the service for customer 6, the drone is launched towards customer 7, while the truck travels towards node 8. At node 8, they must wait for each other before rejoining. Finally, the truck travels from node 8 back to the depot, while the drone needs to serve customer 9 before returning.

The objective of the decision-maker is to determine routes for the truck and the drone to provide services to all customers within specified deadlines as much as possible. To ensure the feasibility and practicality of the model, the following assumptions are imposed on the operations of the truck-and-drone delivery system:

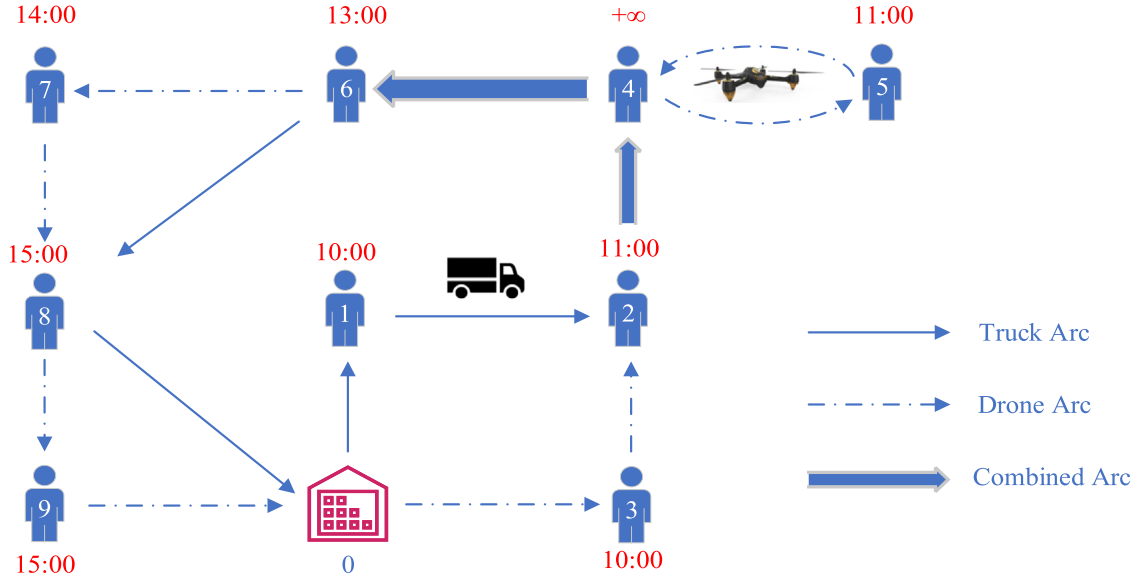


Fig. 2. An illustration example of a TDRP-D with nine customers.

- Due to the capacity limitation, the drone can only serve one customer in a single launch-retrieve process.
- For each customer $i \in \mathcal{N}$ and any arc $(i, j) \in \mathcal{A}$, the drone takes no more time to serve the customer and traverse the arc than the truck, i.e., $s_i^D + z_{ij}^D \leq s_i^T + z_{ij}^T$.
- The drone launching/retrieving is fully automatic, allowing the truck driver to freely travel or serve customers while the drone is launching/retrieving (Tamke and Buscher, 2021).
- The truck can retrieve the drone at the same node where it was launched (Roberti and Ruthmair, 2021). For example, as illustrated in Fig. 2, the truck launches the drone at node 4 to serve customer 5 and waits for it to return at the same node.
- The truck and the drone can also serve their respective customers simultaneously. The truck launches the drone at one node and retrieves it at the next node where the truck is about to arrive (Roberti and Ruthmair, 2021). For instance, the truck traverses the arc (6, 8) to serve customer 8, and, in the meanwhile, the drone traverses the leg $6 \rightarrow 7 \rightarrow 8$ to serve customer 7 in Fig. 2.
- Customers are serviced immediately as either the drone or the truck arrives at the corresponding customer node. If the truck or the drone arrives at node i later than $\bar{\tau}_i$, the service will be deemed poor service level (Zhang et al., 2019, 2021).
- If the drone travels an arc in tandem with the truck, its travel time equals the truck's travel time z_{ij}^T , because, when passing through a combination arc, the drone is not launched, but is transported by the truck. In this case, it is assumed that only the truck provides service to customer i , while the drone does not.
- The drone always has sufficient battery power during a single collaborative delivery task.

These assumptions are realistic and reasonable, with most of them having been applied in the literature.

Finally, for easy reference, we present all the notations in Table 2.

4. Feasible routes for TDRP-D

For each $i, j \in \{0, 0'\} \cup \mathcal{N}$, denote by

$$\delta^+(i) = \{(i, j) \in \mathcal{A} \mid j \in \{0'\} \cup \mathcal{N}, i \neq j\} \quad \text{and} \quad \delta^-(j) = \{(i, j) \in \mathcal{A} \mid i \in \{0\} \cup \mathcal{N}, i \neq j\},$$

the set of outgoing arcs of i and the set of the incoming arcs of j , respectively. For ease of notation, we also refer to them as the sets of adjacent nodes, i.e.,

$$\delta^+(i) = \{j \in \{0'\} \cup \mathcal{N} \mid j \neq i, (i, j) \in \mathcal{A}\} \quad \text{and} \quad \delta^-(j) = \{i \in \{0\} \cup \mathcal{N} \mid i \neq j, (i, j) \in \mathcal{A}\},$$

if no confusion is caused. It is assumed that $\delta^-(0) = \delta^+(0') = \emptyset$.

Table 2
The description of key notations used in the paper.

Type	Notation	Description
Indicies	i, j, l, h	Equivalently used indicies for the nodes (customers and depot)
	a	Index for the arcs
	k	Index for the samples
Sets	$\mathbb{R}_+$	Set of positive real numbers
	$\mathcal{G}$	A complete directed graph
	$\mathcal{V}$	Set of all nodes in the network
	$\mathcal{A}$	Set of all arcs in the network
	$\mathcal{N}$	Set of all customers
	$\overline{\mathcal{N}}$	Set of customers with positive finite deadlines
	$\mathcal{X}$	Set of feasible routing solutions
	$\mathcal{S}$	Set of feasible truck-and-drone flows
	$\delta^+(i)$	Set of outgoing arcs (or nodes) of the node i
	$\delta^-(i)$	Set of incoming arcs (or nodes) of the node i
Parameters	$\mathcal{Z}$	Budget uncertain set
	$\bar{e}_i$	Service deadline of customer i (in seconds)
	s_i^T	Service time of customer i by the truck (in seconds)
	s_i^D	Service time of customer i by the drone (in seconds)
	z_{ij}^T	Travel time of the truck over the arc (i, j) (in seconds)
	z_{ij}^D	Travel time of the drone over the arc (i, j) (in seconds)
	$\hat{z}_k$	The k th sample data of travel time
	M	A specified sufficient large positive number
	K	The number of samples
	$\hat{\mathbb{P}}_K$	The sample-based empirical distribution
Auxiliary variables	Γ	Budget parameter of the uncertain set
	y_i^T	Binary variable: 1 if the truck visits node i alone; 0 otherwise.
	y_i^D	Binary variable: 1 if the drone visits node i alone; 0 otherwise.
	y_i^C	Binary variable: 1 if node i is visited by the truck-and-drone; 0 otherwise.
	ξ_l	Service delay function of node l
Decision variables	ρ	Service level function
	x_{ij}^T	Binary variable: 1 if the truck travels from node i to node j ; 0 otherwise.
	x_{ij}^D	Binary variable: 1 if the drone travels from node i to node j ; 0 otherwise.
	t_i	Service arrival time of node i
	η_i	Service delay value of node i

To characterize the feasible routes for the TDRP-D, we adopt the following truck-and-drone flow set formulation:

$$S = \left\{ \begin{array}{l} \mathbf{x}^T, \mathbf{x}^D \in \{0, 1\}^{|\mathcal{A}|} \\ \mathbf{y}^T, \mathbf{y}^D, \mathbf{y}^C \in \{0, 1\}^{|\mathcal{N}|} \end{array} \left\{ \begin{array}{ll} \sum_{a \in \delta^+(0)} x_a^T = \sum_{a \in \delta^-(0')} x_a^T = 1, & (1a) \\ \sum_{a \in \delta^+(0)} x_a^D = \sum_{a \in \delta^-(0')} x_a^D = 1, & (1b) \\ \sum_{a \in \delta^-(i)} x_a^T = \sum_{a \in \delta^+(i)} x_a^T, & \forall i \in \mathcal{N}, \quad (1c) \\ \sum_{a \in \delta^-(i)} x_a^D = \sum_{a \in \delta^+(i)} x_a^D, & \forall i \in \mathcal{N}, \quad (1d) \\ \sum_{a \in \delta^+(i)} x_a^T = y_i^T + y_i^C, & \forall i \in \mathcal{N}, \quad (1e) \\ \sum_{a \in \delta^+(i)} x_a^D = y_i^D + y_i^C, & \forall i \in \mathcal{N}, \quad (1f) \\ y_i^T + y_i^D + y_i^C = 1, & \forall i \in \mathcal{N}, \quad (1g) \\ x_{ij}^D \leq y_i^C + y_j^C, & \forall i \neq j \in \mathcal{N}, \quad (1h) \\ x_a^T, x_a^D \in \{0, 1\}, & \forall a \in \mathcal{A}, \quad (1i) \\ y_i^T, y_i^D, y_i^C \in \{0, 1\}, & \forall i \in \mathcal{N}. \quad (1j) \end{array} \right. \right\}. \quad (1)$$

For an arc $a \in \mathcal{A}$, if it is traversed by the truck alone, it is termed a *truck arc*, denoted by $x_a^T = 1$ and $x_a^D = 0$. Similarly, if the arc $a \in \mathcal{A}$ is traversed by the drone alone, it is referred to as a *drone arc*, with $x_a^D = 1$ and $x_a^T = 0$. Moreover, if an arc $a \in \mathcal{A}$ is traversed by the truck in tandem with the drone, it is termed a *combined arc*, with $x_a^T = x_a^D = 1$. A sequence of truck arcs (drone arcs, combined arcs) forms a *truck leg* (*drone leg*, *combined leg*). For instance, in the routing flow illustrated in Fig. 3, $i \rightarrow j \rightarrow k$ is a truck leg, $i \rightarrow l \rightarrow k$ is a drone leg, and $k \rightarrow h$ is a combined leg. For each customer node $i \in \mathcal{N}$, the variable y_i^T is set to 1 if node i is visited by the truck alone; otherwise, it is set to 0. Similarly, the variable $y_i^D = 1$ if node i is visited by the drone alone; otherwise it equals 0.

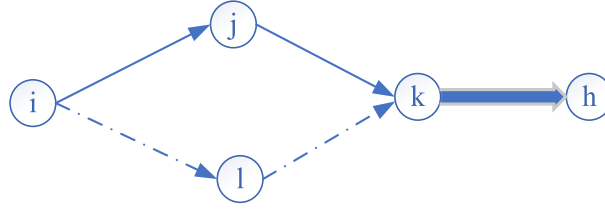


Fig. 3. Truck arc, drone arc and combined arc.

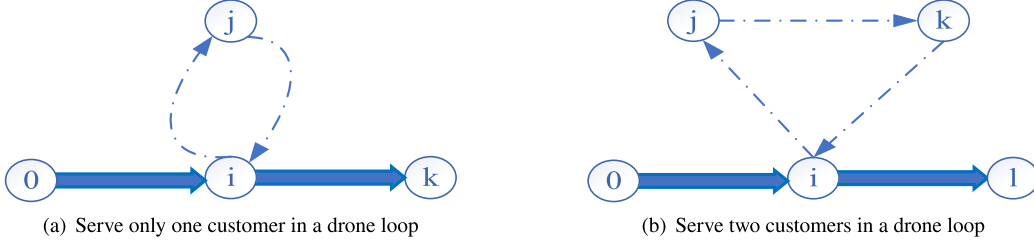


Fig. 4. Drone loops.

Additionally, $y_i^C = 1$ if node i is visited by both the truck and the drone. In the illustrated routing in Fig. 3, $y_i^C = y_k^C = y_h^C = 1$, $y_j^T = 1$, and $y_l^D = 1$.

Next, we explain the constraints formulated in S one by one. The constraints (1a) and (1b) ensure that both the truck and the drone leave and return to the depot exactly once. The constraints (1c) and (1d) state the flow conservation laws for the truck and the drone, respectively. The constraints (1e) and (1f) link the variables x^T and x^D with the variables y^T , y^D , y^C . If a node i is passed by the truck, then $\sum_{a \in \delta^+(i)} x_a^T = 1$, in which case $y_i^T + y_i^C = 1$ always holds. This is because if the truck passes the node alone, $y_i^T = 1$, $y_i^C = 0$; if it passes the node in tandem with the drone, $y_i^T = 0$, $y_i^C = 1$. The constraint (1g) ensures that each node is visited exactly once by the truck alone, or by the drone alone, or by the truck-and-drone. The constraint (1h) ensures that the drone can only serve one customer in each launch-retrieve loop. For example, in Fig. 4(a), $x_{ij}^D = 1$ is upper bounded by $y_i^C + y_j^C = 1$. In contrast, in Fig. 4(b), $x_{jk}^D = 1$ cannot be upper bounded by $y_j^C + y_k^C = 0$, leading to the exclusion of this drone loop by constraint (1h). Finally, the constraints (1i) and (1j) describe the range of the variables.

We remark that, the set (1) is built upon the compact formulation proposed in Roberti and Ruthmair (2021), with two modifications made. First, the constraint (1h) is slightly different from its counterpart used in Roberti and Ruthmair (2021), with the later being $x_{ij}^D + x_{ji}^D \leq y_i^C + y_j^C$. The constraint $x_{ij}^D + x_{ji}^D \leq y_i^C + y_j^C$ ensures the drone traverse an arc (i, j) if and only if the truck visits at least one of the two nodes, the constraint (1h) can also achieve this. However, the constraint $x_{ij}^D + x_{ji}^D \leq y_i^C + y_j^C$ rules out the scenario that the drone is launched from node i to serve a customer and then returns to the same node i , as illustrated in Fig. 4(a), because in this case, $x_{ij}^D + x_{ji}^D = 2 > y_i^C + y_j^C = 1$. Our revised constraint $x_{ij}^D \leq y_i^C + y_j^C$ is also true in this case. Second, we remove the constraint $x_{0i}^D + x_{i0'}^D \leq 1, \forall i \in \mathcal{N}$, which rules out the drone loops rooted at the depot, i.e., $0 \rightarrow i \rightarrow 0'$ for any $i \in \mathcal{N}$. It may be feasible and even optimal in certain practical scenarios.

Building upon the previous set formulation of truck-and-drone flows, we define the set of feasible routes for the TDRP-D as:

$$\mathcal{X} = \left\{ (x^T, x^D) \mid (x^T, x^D, y^T, y^D, y^C) \in S \text{ for some } y^T, y^D, y^C \in \{0, 1\}^{|\mathcal{N}|} \right\}. \quad (2)$$

One can see that, there is one-to-one correspondence between $\mathcal{X}$ and S .

5. Model formulations

For the TDRP-D, the uncertainty of travel time can be influenced by various factors such as traffic congestion, weather conditions, drone payload capacity, and more. This uncertainty necessitates the consideration of uncertain travel time in the research.

In practice, the exact distribution of travel time are often unknown. However, it is possible to collect a finite number of sample data for travel time. This uncertainty can be addressed by leveraging information from these samples. Two scenarios are typically encountered in the real world: (i) when the sample size is large enough to approximate the true distribution well through a sample-based empirical distribution, and (ii) when the sample size is too small to extract abundant statistical information. In what follows, we refer to the first and second scenarios as the “big data” and “small data” cases, respectively.

This study addresses the TDRP-D problems in these two scenarios separately. In the case of big data, the problem is formulated using a SAA approach commonly employed in studying stochastic programming problems. In the case of small data, an uncertain set based on the samples is formulated, and a RO model for the TDRP-D is established.

5.1. The STDRP-D model

Let $\mathbf{z} := (\mathbf{z}^T, \mathbf{z}^D) \in \mathbb{R}_+^{|\mathcal{A}|} \times \mathbb{R}_+^{|\mathcal{A}|}$ be a deterministic tuple of travel time of the truck-and-drone. Suppose both the truck and the drone depart from the depot at $t_0 = 0$ simultaneously. Given a feasible routing solution $\mathbf{x} := (\mathbf{x}^T, \mathbf{x}^D) \in \mathcal{X}$, for each $l \in \mathcal{N}$, the service arrival time at node l can be recursively determined by

$$t_l(\mathbf{x}, \mathbf{z}) = \max_{i \in \delta^-(l)} \{t_i + (s_i^T + z_{il}^T) x_{il}^T, t_i + (s_i^D + z_{il}^D) x_{il}^D\}, \quad (3)$$

In this equation, the maximization is taken over all arcs (i, l) along the routes $\mathbf{x}^T, \mathbf{x}^D$. For a truck arc (i, l) , i.e., $x_{il}^T = 1, x_{il}^D = 0$, the truck departure time at node i is $t_i + s_i^T$, plusing the travel time of the arc (i, l) , yields the service arrival time at node l . Hence, $t_l = t_i + s_i^T + z_{il}^T$. Similarly, for a drone arc (i, l) where, $x_{il}^T = 0, x_{il}^D = 1$, then $t_l = t_i + s_i^D + z_{il}^D$. If (i, l) is a combined arc, i.e., $x_{il}^T = x_{il}^D = 1$, by Assumption (g), only the truck provides service to customer in this case, and the travel time is z_{il}^T . Thus, $t_l = \max\{t_i + s_i^T + z_{il}^T, t_i + s_i^D + z_{il}^D\}$.

Given a routing solution $\mathbf{x} \in \mathcal{X}$ and a deterministic travel time $\mathbf{z}$, the lateness magnitude of service starting at each node $l \in \mathcal{N}$ can be characterized by the following *delay function*:

$$\xi_l(\mathbf{x}, \mathbf{z}) = (t_l(\mathbf{x}, \mathbf{z}) - \bar{t}_l)^+, \quad (4)$$

where $(\cdot)^+ := \max(\cdot, 0)$. The service for customer l is late with respect to the deadline if and only if $\xi_l(\mathbf{x}, \mathbf{z}) > 0$. Denote by $\xi(\mathbf{x}, \mathbf{z}) := (\xi_l(\mathbf{x}, \mathbf{z}))_{l \in \mathcal{N}}$ the vector of delays. The delivery service level can be measured by a function of $\xi(\mathbf{x}, \mathbf{z})$, e.g., $\rho(\mathbf{x}, \mathbf{z}) = \sum_{l \in \mathcal{N}} \xi_l(\mathbf{x}, \mathbf{z})$ or $\rho(\mathbf{x}, \mathbf{z}) = \max_{l \in \mathcal{N}} \xi_l(\mathbf{x}, \mathbf{z})$, which measure the total lateness magnitude and the maximum delay value, respectively.

For instance, if we take $\rho(\mathbf{x}, \mathbf{z}) = \sum_{l \in \mathcal{N}} \xi_l(\mathbf{x}, \mathbf{z})$ as the service delay risk index. As the service starting time at each node can be recursively determined, the service delay risk index $\rho(\mathbf{x}, \mathbf{z})$ can be computed by solving the following linear programming:

$$\min \sum_{l \in \mathcal{N}} (t_l - \bar{t}_l)^+ \quad (5a)$$

$$\text{s.t. } t_i + s_i^T + z_{ij}^T \leq t_j + M(1 - x_{ij}^T), \quad \forall (i, j) \in \mathcal{A}, \quad (5b)$$

$$t_i + s_i^D + z_{ij}^D \leq t_j + M(1 - x_{ij}^D), \quad \forall (i, j) \in \mathcal{A}. \quad (5c)$$

where M is a specified sufficient large positive number.

Suppose one has access to a finite number of travel time samples, $\{\hat{\mathbf{z}}_1, \hat{\mathbf{z}}_2, \dots, \hat{\mathbf{z}}_K\}$. The SAA method assigns equal weight to each sample and constructs an empirical distribution from the samples as follows:

$$\hat{\mathbb{P}}_K = \frac{1}{K} \sum_{k=1}^K \delta_{\hat{\mathbf{z}}_k}, \quad (6)$$

where $\delta_{\hat{\mathbf{z}}_k}$ is the Dirac distribution at $\hat{\mathbf{z}}_k$. Then, $\hat{\mathbb{P}}_K(\tilde{\mathbf{z}} = \hat{\mathbf{z}}_k) = \frac{1}{K}$ for each $k = 1, \dots, K$, and $\hat{\mathbb{P}}_K(\tilde{\mathbf{z}} = \mathbf{z}') = 0$ for any $\mathbf{z}'$ outside the set of samples.

The SAA for the truck-and-drone routing problem with deadlines (STDPR-D) aims to find a feasible routing solution that minimizes the service delay risk under the empirical distribution $\hat{\mathbb{P}}_K$. That is, it solves the following stochastic programming:

$$\min_{\mathbf{x} \in \mathcal{X}} \mathbb{E}_{\hat{\mathbb{P}}_K} [\rho(\mathbf{x}, \tilde{\mathbf{z}})] = \min_{\mathbf{x} \in \mathcal{X}} \frac{1}{K} \sum_{k=1}^K \rho(\mathbf{x}, \hat{\mathbf{z}}_k). \quad (7)$$

By the reformulation (5), the min-sum STDPR-TW is equivalent to:

$$\min_{\mathbf{x}, \{\eta^k, t^k\}_{k=1, \dots, K}} \frac{1}{K} \sum_{k=1}^K \sum_{l \in \mathcal{N}} \eta_l^k \quad (8a)$$

$$\text{s.t. } t_i^k + s_i^T + \hat{z}_{k,ij}^T \leq t_j^k + M(1 - x_{ij}^T), \quad \forall (i, j) \in \mathcal{A}, \forall k = 1, \dots, K, \quad (8b)$$

$$t_i^k + s_i^D + \hat{z}_{k,ij}^D \leq t_j^k + M(1 - x_{ij}^D), \quad \forall (i, j) \in \mathcal{A}, \forall k = 1, \dots, K, \quad (8c)$$

$$\eta_l^k \geq t_l^k - \bar{t}_l, \quad \forall l \in \mathcal{N}, \forall k = 1, \dots, K, \quad (8d)$$

$$\eta_l^k \geq 0, \quad \forall l \in \mathcal{N}, \forall k = 1, \dots, K, \quad (8e)$$

$$\mathbf{x} \in \mathcal{X}. \quad (8f)$$

This problem is a LP with binary variables. For smaller instances, direct solutions using optimization solvers like CPLEX and GUROBI are feasible. However, when the problem size is large, attempting to solve it directly with these solvers becomes impractical within acceptable time limits.

Remark 5.1. The model with the objective function $\rho(\mathbf{x}, \mathbf{z}) = \max_{l \in \mathcal{N}} \xi_l(\mathbf{x}, \mathbf{z})$ can be established in a similar way: $\min_{\mathbf{x} \in \mathcal{X}} \max_{l \in \mathcal{N}} \xi_l(\mathbf{x}, \mathbf{z})$, which is referred to the min-max STDRP-D model. Similar to (8), the min-max STDRP-D model can be approximated by

the following LP:

$$\min_{\mathbf{x}, \{\alpha^k, t^k\}_{k=1, \dots, K}} \frac{1}{K} \sum_{k=1}^K \alpha^k \quad (9a)$$

$$\text{s.t. } t_i^k + s_i^T + \hat{z}_{k,ij}^T \leq t_j^k + M(1 - x_{ij}^T), \quad \forall (i, j) \in \mathcal{A}, \forall k = 1, \dots, K, \quad (9b)$$

$$t_i^k + s_i^D + \hat{z}_{k,ij}^D \leq t_j^k + M(1 - x_{ij}^D), \quad \forall (i, j) \in \mathcal{A}, \forall k = 1, \dots, K, \quad (9c)$$

$$\alpha^k \geq t_l^k - \bar{t}_l, \quad \forall l \in \bar{\mathcal{N}}, \forall k = 1, \dots, K, \quad (9d)$$

$$\alpha^k \geq 0, \quad \forall k = 1, \dots, K, \quad (9e)$$

$$\mathbf{x} \in \mathcal{X}, \quad (9f)$$

Since both the objective functions of the min-sum and min-max models are linear, the solving procedure proposed later for the min-sum model also applies to solving the min-max model. Therefore, we mainly focus on analyzing the min-sum model in the following context.

5.2. The RTDRP-D model

In the scenario of limited data, where the empirical distribution $\hat{\mathbb{P}}_K$ may deviate significantly from the true distribution of travel time. There is a high probability that the STDPR-D model yields a routing solution with poor performance.

To mitigate this challenge, we make an assumption that the travel time adhere to a specific uncertain set and employ the RO framework to handle the uncertainty. In this context, we adopt the commonly used budget uncertain set (Bertsimas and Sim, 2003) to model the uncertainty of travel time, namely,

$$\mathcal{Z} \equiv \left\{ \mathbf{z} \in \mathbb{R}^{|\mathcal{A}| \times |\mathcal{A}|} \mid z_a^T \in [z_a^T, z_a^T + \hat{z}_a^T], z_a^D \in [z_a^D, z_a^D + \hat{z}_a^D], \forall a \in \mathcal{A}, \sum_{a \in \mathcal{A}} \left[\frac{z_a^T - z_a^T}{\hat{z}_a^T} + \frac{z_a^D - z_a^D}{\hat{z}_a^D} \right] \leq \Gamma \right\}, \quad (10)$$

where $\Gamma > 0$ be a given budget parameter. For each arc $a \in \mathcal{A}$, z_a^T and z_a^D are the nominal values of the truck travel time and the drone travel time over the arc, respectively. $\hat{z}_a^T$ and $\hat{z}_a^D$ are the maximum possible deviations from the nominal travel time. The intervals $[z_a^T, z_a^T + \hat{z}_a^T]$ and $[z_a^D, z_a^D + \hat{z}_a^D]$ are respectively the ranges of the uncertain travel time z_a^T and z_a^D . The inequality $\sum_{a \in \mathcal{A}} \left[\frac{z_a^T - z_a^T}{\hat{z}_a^T} + \frac{z_a^D - z_a^D}{\hat{z}_a^D} \right] \leq \Gamma$ is used to control the total deviation of all travel time from their nominal values. The parameter $\Gamma > 0$ is termed as the “budget of uncertainty”. Obviously, $\sum_{a \in \mathcal{A}} \left[\frac{z_a^T - z_a^T}{\hat{z}_a^T} + \frac{z_a^D - z_a^D}{\hat{z}_a^D} \right] \leq 2|\mathcal{A}|$ always holds, thus, it is assumed that $\Gamma \in [0, 2|\mathcal{A}|]$.

Particularly, when $\Gamma = 0$, the uncertain set $\mathcal{Z}$ shrinks to the singleton $\mathcal{Z} = \{(z_a^T, z_a^D)_{a \in \mathcal{A}}\}$, in which case all the travel times are deterministic.

All the parameters in (10) can be estimated from the samples. For instance, for each arc $a \in \mathcal{A}$, we may set

$$z_a^T = \min_{k=1, \dots, K} \hat{z}_{k,a}^T, \quad z_a^D = \min_{k=1, \dots, K} \hat{z}_{k,a}^D, \quad \hat{z}_a^T = \max_{k=1, \dots, K} (\hat{z}_{k,a}^T - z_a^T), \quad \hat{z}_a^D = \max_{k=1, \dots, K} (\hat{z}_{k,a}^D - z_a^D). \quad (11)$$

In this approach, the lower bound of the travel time for an arc is determined by selecting the minimum value from the travel time samples associated with that arc. Simultaneously, the length of the uncertain interval is defined by considering the largest deviation among the samples for that particular arc. The budget parameter Γ can be estimated from the samples, e.g., by setting,

$$\Gamma = \left\lceil \frac{1}{K} \sum_{k=1}^K \sum_{a \in \mathcal{A}} \left[\frac{\hat{z}_{k,a}^T - z_a^T}{\hat{z}_a^T} + \frac{\hat{z}_{k,a}^D - z_a^D}{\hat{z}_a^D} \right] \right\rceil, \quad (12)$$

where $\lceil \cdot \rceil$ is the upward rounding function. The formula (12) estimates the budget by evaluating the mean of the total deviation values of the system.

The robust truck-and-drone routing problem with deadlines (RTDRP-D) aims to find a feasible routing solution that minimizes the worst-case value of service delay risk over the uncertain set. That is, it solves the RO problem:

$$\min_{\mathbf{x} \in \mathcal{X}} \max_{\mathbf{z} \in \mathcal{Z}} \rho(\mathbf{x}, \mathbf{z}). \quad (13)$$

By (5), we reformulate the RTDRP-D as:

$$\min_{\mathbf{x} \in \mathcal{X}} \max_{\mathbf{z} \in \mathcal{Z}} \min_{\eta, t} \sum_{l \in \mathcal{N}} \eta_l \quad (14a)$$

$$\text{s.t. } t_i + s_i^T + z_{ij}^T \leq t_j + M(1 - x_{ij}^T), \quad \forall (i, j) \in \mathcal{A}, \quad (14b)$$

$$t_i + s_i^D + z_{ij}^D \leq t_j + M(1 - x_{ij}^D), \quad \forall (i, j) \in \mathcal{A}, \quad (14c)$$

$$\eta_l \geq t_l - \bar{t}_l, \quad \forall l \in \bar{\mathcal{N}}, \quad (14d)$$

$$\eta_l \geq 0, \quad \forall l \in \bar{\mathcal{N}}. \quad (14e)$$

It is a two-stage adaptive RO problem which is NP-hard in general (Bertsimas and de Ruiter, 2016). The first-stage problem is to decide an optimal routing solution $\mathbf{x}$. The second-stage problem is to derive the worst-case service delay risk under $\mathbf{x}$.

6. Solving procedures

This section develops Benders Decomposition (BD) approaches to solve the proposed STDRP-D and the RTDRP-D models. We solve the STDRP-D in Section 6.1. Then, we present a dual formulation for the second stage problem of (14) in Section 6.2. Then, we develop a BD algorithm to solve the RTDRP-D in Section 6.3. Finally, we establish some valid inequalities that can boost the computational efficiency of the algorithms in Section 6.4.

6.1. Benders decomposition for the STDRP-D

In the big data case, problem (8) cannot be directly solved by commercial solvers within an acceptable time in general. The classic SAA routing problems can be more efficiently solved by the BD approach (Adulyasak and Jaillet, 2016; Zhang et al., 2019). In this study, we extend the approach to the SAA truck-and-drone routing problem.

By rearranging the objective function of problem (8), the STDRP-D can be reexpressed as follows:

$$\min_{\mathbf{x}, \alpha} \sum_{l \in \bar{\mathcal{N}}} \alpha_l \quad (15a)$$

$$\text{s.t. } \alpha_l \geq F_l(\mathbf{x}), \quad (15b)$$

$$\mathbf{x} \in \mathcal{X}, \quad (15c)$$

where the function $F_l(\mathbf{x}) := \frac{1}{K} \sum_{k=1}^K \xi_l^k(\mathbf{x}, \mathbf{z}_k)$ represents the service delay risk at node $l \in \bar{\mathcal{N}}$ over the samples.

Let $0 \rightarrow i_1^T \rightarrow \dots \rightarrow i_{m_l}^T$ and $0 \rightarrow i_1^D \rightarrow \dots \rightarrow i_{n_l}^D$ the truck and drone routing paths towards l along $\mathbf{x}$, respectively. Note that $i_{m_l}^T$ or $i_{n_l}^D$ may not equal l when the node is traversed by the truck or the drone alone. Denote by $\mathcal{A}_l^T(\mathbf{x}) = \{(0, i_1^T), (i_1^T, i_2^T), \dots, (i_{m_l-1}^T, i_{m_l}^T)\}$ and $\mathcal{A}_l^D(\mathbf{x}) = \{(0, i_1^D), (i_1^D, i_2^D), \dots, (i_{n_l-1}^D, i_{n_l}^D)\}$. We also use the notations $\mathcal{A}_l^T$ and $\mathcal{A}_l^D$ for simplicity, when no confusion is caused.

Then, the service delay risk $F_l(\mathbf{x})$ can be reformulated as :

$$F_l(\mathbf{x}) = \min_{\eta, t} \frac{1}{K} \sum_{k=1}^K \eta_l^k \quad (16a)$$

$$\text{s.t. } t_i^k + s_i^T + \hat{z}_{k,ij}^T \leq t_j^k, \quad \forall (i, j) \in \mathcal{A}_l^T, \forall k = 1, 2, \dots, K, \quad (16b)$$

$$t_i^k + s_i^D + \hat{z}_{k,ij}^D \leq t_j^k, \quad \forall (i, j) \in \mathcal{A}_l^D, \forall k = 1, 2, \dots, K, \quad (16c)$$

$$\eta_l^k \geq t_l^k - \bar{\tau}_l, \quad \forall k = 1, 2, \dots, K, \quad (16d)$$

$$\eta_l^k \geq 0, \quad \forall k = 1, 2, \dots, K. \quad (16e)$$

which is a linear programming, and thus the strong duality holds. Its dual problem is

$$\max_{\lambda_l, \psi_l} \sum_{k=1}^K \sum_{(i,j) \in \mathcal{A}} \left(x_{ij}^T \hat{z}_{k,ij}^T \lambda_{l,ij}^{T,k} + x_{ij}^D \hat{z}_{k,ij}^D \lambda_{l,ij}^{D,k} \right) - \sum_{k=1}^K \psi_l^k \bar{\tau}_l \quad (17a)$$

$$\text{s.t. } \psi_l^k - \sum_{j \in \delta^-(l)} \left(x_{jl}^T \lambda_{l,jl}^{T,k} + x_{jl}^D \lambda_{l,jl}^{D,k} \right) \geq 0, \quad \forall k = 1, 2, \dots, K, \quad (17b)$$

$$\sum_{(i,j) \in \mathcal{A}_l^T} \lambda_{l,ij}^{T,k} + \sum_{(i,j) \in \mathcal{A}_l^D} \lambda_{l,ij}^{D,k} - \sum_{(j,i) \in \mathcal{A}_l^T} \lambda_{l,ji}^{T,k} - \sum_{(j,i) \in \mathcal{A}_l^D} \lambda_{l,ji}^{D,k} \geq 0, \quad \forall i \in \mathcal{V} \setminus \{l\}, \forall k = 1, 2, \dots, K, \quad (17c)$$

$$\lambda_{l,ij}^{T,k}, \lambda_{l,ij}^{D,k} \geq 0, \quad \forall (i, j) \in \mathcal{A}, \forall k = 1, 2, \dots, K \quad (17d)$$

$$\lambda_{l,ij}^{T,k} = 0, \quad \forall (i, j) \notin \mathcal{A}_l^T, \forall k = 1, 2, \dots, K \quad (17e)$$

$$\lambda_{l,ij}^{D,k} = 0, \quad \forall (i, j) \notin \mathcal{A}_l^D, \forall k = 1, 2, \dots, K \quad (17f)$$

$$0 \leq \psi_l^k \leq \frac{1}{K}, \quad \forall k = 1, 2, \dots, K. \quad (17g)$$

Obviously, the problem is bounded, and the optimal solution (λ_l, ψ_l) provides an optimality cut to the original problem:

$$\alpha_l \geq F_l(\mathbf{x}) = \sum_{k=1}^K \sum_{(i,j) \in \mathcal{A}} \left(x_{ij}^T \hat{z}_{k,ij}^T \lambda_{l,ij}^{T,k} + x_{ij}^D \hat{z}_{k,ij}^D \lambda_{l,ij}^{D,k} \right) - \sum_{k=1}^K \psi_l^k \bar{\tau}_l. \quad (18)$$

Given a set $\mathcal{O}_l = \{(\lambda_l, \psi_l)\}$ for each $l \in \bar{\mathcal{N}}$, we can define the corresponding restricted master problem (RMP) of (8) as follows:

$$\min_{\mathbf{x}, \alpha, \eta, t} \sum_{l \in \bar{\mathcal{N}}} \alpha_l \quad (19a)$$

$$\begin{aligned}
\text{s.t. } \alpha_l &\geq \sum_{k=1}^K \sum_{(i,j) \in \mathcal{A}} \left(x_{ij}^T \hat{z}_{k,ij}^T \lambda_{l,ij}^{T,k} + x_{ij}^D \hat{z}_{k,ij}^D \lambda_{l,ij}^{D,k} \right) - \sum_{k=1}^K \psi_l^k \bar{\tau}_l, & \forall (\lambda_l, \psi_l) \in \mathcal{O}_l, \forall l \in \overline{\mathcal{N}}, & (19b) \\
t_i^k + s_i^T + \hat{z}_{k,ij}^T &\leq t_j^k + M(1 - x_{ij}^T), & \forall (i, j) \in \mathcal{A}, \forall k = 1, \dots, K, & (19c) \\
t_i^k + s_i^D + \hat{z}_{k,ij}^D &\leq t_j^k + M(1 - x_{ij}^D), & \forall (i, j) \in \mathcal{A}, \forall k = 1, \dots, K, & (19d) \\
\eta_l^k &\geq t_l^k - \bar{\tau}_l, & \forall l \in \overline{\mathcal{N}}, \forall k = 1, \dots, K, & (19e) \\
\eta_l^k &\geq 0, & \forall l \in \overline{\mathcal{N}}, \forall k = 1, \dots, K, & (19f) \\
\mathbf{x} &\in \mathcal{X}. & & (19g)
\end{aligned}$$

Algorithm 1 Benders Decomposition for the STDRP-D model.

Initialization:

For any $l \in \overline{\mathcal{N}}$, initialize the set of optimality cut pool $\mathcal{O}_l = \emptyset$.

Iterations:

For iteration $m \geq 1$,

1: Solve the MP (19), let $(\mathbf{x}_m, \alpha_m)$ be an optimal solution.

2: For any $l \in \overline{\mathcal{N}}$, solving the subproblem formulated in (17), denote by $(\lambda_{m,l}, \pi_{m,l})$ the optimum. Add the optimality cut

$$\alpha_l \geq \sum_{k=1}^K \sum_{(i,j) \in \mathcal{A}} \left(x_{ij}^T \hat{z}_{k,ij}^T \lambda_{l,ij}^{T,k} + x_{ij}^D \hat{z}_{k,ij}^D \lambda_{l,ij}^{D,k} \right) - \sum_{k=1}^K \psi_l^k \bar{\tau}_l$$

to the cut pool $\mathcal{O}_l$ of the master problem, i.e., update $\mathcal{O}_l := \mathcal{O}_l \cup \{(\lambda_{m,l}, \pi_{m,l})\}$.

3: If $F_l(\mathbf{x}_m) = \alpha_{m,l}$ for any $l \in \overline{\mathcal{N}}$, then stop the algorithm and return $\mathbf{x}_m$ as the optimal routing solution. Otherwise, update $m := m + 1$ and go to Step 1.

Algorithm 1 outlines the BD algorithm for solving the STDRP-D model. This iterative approach employs cutting-plane operations to refine the solution. In each iteration, the dual subproblem (17) is addressed only after achieving optimality in solving the RMP (19).

The selection of the initial feasible routing solution $\mathbf{x}_1 \in \mathcal{X}$ in Algorithm 1 significantly influences the computational efficiency of the BD algorithm. We can choose an appropriate initial routing solution by solving a sample-based deterministic optimization problem. For each $(i, j) \in \mathcal{A}$,

$$z_{ij}^{T,\dagger} := \frac{1}{K} \sum_{k=1}^K \hat{z}_{k,ij}^T, \quad z_{ij}^{D,\dagger} := \frac{1}{K} \sum_{k=1}^K \hat{z}_{k,ij}^D, \quad (20)$$

and $\mathbf{z}^\dagger = (z_{ij}^{T,\dagger}, z_{ij}^{D,\dagger})_{(i,j) \in \mathcal{A}}$. By replacing the uncertain travel time z_{ij}^T, z_{ij}^D by $z_{ij}^{T,\dagger}, z_{ij}^{D,\dagger}$, we consider the deterministic TDRP-D problem $\min_{\mathbf{x} \in \mathcal{X}} \rho(\mathbf{x}, \mathbf{z}^\dagger)$, that is,

$$\min_{\mathbf{x}, \eta, t} \sum_{l \in \overline{\mathcal{N}}} \eta_l \quad (21a)$$

$$\text{s.t. } t_i + s_i^T + z_{ij}^{T,\dagger} \leq t_j + M(1 - x_{ij}^T), \quad \forall (i, j) \in \mathcal{A}, \quad (21b)$$

$$t_i + s_i^D + z_{ij}^{D,\dagger} \leq t_j + M(1 - x_{ij}^D), \quad \forall (i, j) \in \mathcal{A}, \quad (21c)$$

$$\eta_l \geq t_l - \bar{\tau}_l, \quad \forall l \in \overline{\mathcal{N}}, \quad (21d)$$

$$\eta_l \geq 0, \quad \forall l \in \overline{\mathcal{N}}, \quad (21e)$$

$$\mathbf{x} \in \mathcal{X}, \quad (21f)$$

where $M > 0$ is a sufficiently large number. The problem is a LP problem with a number of variables independent of the sample size, making it suitable for direct solution by commercial solvers in general. Let $(\mathbf{x}^\dagger, \eta^\dagger, t^\dagger)$ be an optimal solution to problem (21), we adopt $\mathbf{x}^\dagger$ as the initial routing solution for BD Algorithm 1.

6.2. Dual formulation of the RTDRP-D

Before presenting the detailed BD algorithm to solve the RTDRP-D, it is useful to write out the dual of the second stage problem of (13). For a given feasible routing solution $\mathbf{x}$, the worst-case service delay risk is given by

$$\rho(\mathbf{x}) := \max_{\mathbf{z} \in \mathcal{Z}} \rho(\mathbf{x}, \mathbf{z}) = \max_{\mathbf{z} \in \mathcal{Z}} \min_{\eta, l} \sum_{l \in \bar{\mathcal{N}}} \eta_l, \quad (22)$$

with constraints expressed in problem (14). In this subsection and the following Section 6.3, for the simplicity of mathematical deducing, we let $s_i^T = s_i^D = 0$ for any $i \in \mathcal{N}$, otherwise, we can replace z_{ij}^T, z_{ij}^D by $s_i^T + z_{ij}^T, s_i^D + z_{ij}^D$, respectively.

First, we write out the dual of the inner problem $\rho(\mathbf{x}, \mathbf{z}) = \min_{\eta, l} \sum_{l \in \bar{\mathcal{N}}} \eta_l$. Let $\lambda_{ij}^T, \lambda_{ij}^D, (i, j) \in \mathcal{A}$, and ψ_l ($l \in \bar{\mathcal{N}}$), be the dual variables associated with the first three constraints, respectively. Moreover, for any $(i, j) \in \mathcal{A}$, if $x_{ij}^T = 0$ or $x_{ij}^D = 0$, we impose $\lambda_{ij}^T = 0$ or $\lambda_{ij}^D = 0$, respectively; in addition, for any $l' \notin \bar{\mathcal{N}}$, we impose that $\psi_{l'} = 0$. Then,

$$\rho(\mathbf{x}, \mathbf{z}) = \max_{\lambda, \psi} \sum_{(i,j) \in \mathcal{A}} \left(x_{ij}^T z_{ij}^T \lambda_{ij}^T + x_{ij}^D z_{ij}^D \lambda_{ij}^D \right) - \sum_{l \in \bar{\mathcal{N}}} \psi_l \bar{\tau}_l \quad (23a)$$

$$\text{s.t. } (\lambda, \psi) \in \Omega(\mathbf{x}) \quad (23b)$$

where

$$\Omega(\mathbf{x}) := \left\{ (\lambda, \psi) \left| \begin{array}{ll} \sum_{j \in \delta^+(i)} \left(x_{ij}^T \lambda_{ij}^T + x_{ij}^D \lambda_{ij}^D \right) - \sum_{j \in \delta^-(i)} \left(x_{ji}^T \lambda_{ji}^T + x_{ji}^D \lambda_{ji}^D \right) + \psi_i \geq 0, & \forall i \in \mathcal{N} \cup \{0'\}, \\ \lambda_{ij}^T \geq 0, & \forall (i, j) \in \mathcal{A}, \\ \lambda_{ij}^D \geq 0, & \forall (i, j) \in \mathcal{A}, \\ 0 \leq \psi_l \leq 1, & \forall l \in \bar{\mathcal{N}}, \\ \psi_l = 0, & \forall l \notin \bar{\mathcal{N}}, \end{array} \right. \right\} \quad (24)$$

be the polyhedral defined by the constraints of variables (λ, ψ) .

Second, for each $a \in \mathcal{A}$ and a realized tuple of travel time $\mathbf{z}_a = (z_a^T, z_a^D) \in \mathcal{Z}$, z_a^T and z_a^D can be expressed as

$$z_a^T = \bar{z}_a^T + \delta_a^T \bar{z}_a^T, \quad z_a^D = \bar{z}_a^D + \delta_a^D \bar{z}_a^D \quad (25)$$

with $\delta_a^T, \delta_a^D \in [0, 1]$. To simplify the analysis, we introduce the following uncertain set

$$\Theta \equiv \left\{ \delta \in \mathbb{R}^{|\mathcal{A}| \times |\mathcal{A}|} \left| \delta_a^T, \delta_a^D \in [0, 1], \forall a \in \mathcal{A}, \sum_{a \in \mathcal{A}} (\delta_a^T + \delta_a^D) \leq \Gamma \right. \right\}. \quad (26)$$

The identities (25) establish an one-to-one correspondence between Θ and $\mathcal{Z}$. Therefore, $\rho(\mathbf{x})$ can be reformulated as

$$\rho(\mathbf{x}) = \max_{\delta \in \Theta, (\lambda, \psi) \in \Omega(\mathbf{x})} \sum_{(i,j) \in \mathcal{A}} \left(x_{ij}^T \bar{z}_{ij}^T \lambda_{ij}^T + x_{ij}^D \bar{z}_{ij}^D \lambda_{ij}^D + x_{ij}^T \delta_{ij}^T \bar{z}_{ij}^T \lambda_{ij}^T + x_{ij}^D \delta_{ij}^D \bar{z}_{ij}^D \lambda_{ij}^D \right) - \sum_{l \in \bar{\mathcal{N}}} \psi_l \bar{\tau}_l. \quad (27)$$

which is a bilinear optimization problem. Several approaches have been proposed in the literature to solve bilinear optimization problems, including outer approximation algorithms (Duran and Grossmann, 1986; Fletcher and Leyffer, 1994; Minoux, 2011), heuristic methods (Bertsimas et al., 2013), and so on.

Thanks to the well-organized structure of the uncertain set (26), we can reformulate problem (27) as a MILP problem. When the uncertainty budget Γ is an integer, there exists an optimal solution $(\delta^*, \lambda^*, \psi^*)$ of (27) such that $\delta_a^{T,*}, \delta_a^{D,*} \in \{0, 1\}$ for each arc $a \in \mathcal{A}$ (Gabrel et al., 2014). As a result, the optimal value of z_a^T is either equal to the nominal value $\bar{z}_a^T$, or the worst-case value $\bar{z}_a^T + \bar{z}_a^T$, so as for z_a^D . In what follows, we assume that the budget Γ is an integer. Then, we can replace the constraints $\delta_a^T, \delta_a^D \in [0, 1]$ by $\delta_a^T, \delta_a^D \in \{0, 1\}$, and rewrite problem (27) as an MILP in the following proposition.

Proposition 6.1. *The bilinear optimization problem (27) is equivalent to the MILP problem:*

$$\max_{\delta, \lambda, \psi} \sum_{(i,j) \in \mathcal{A}} \left(x_{ij}^T \bar{z}_{ij}^T \lambda_{ij}^T + x_{ij}^D \bar{z}_{ij}^D \lambda_{ij}^D + x_{ij}^T \delta_{ij}^T \bar{z}_{ij}^T \lambda_{ij}^T + x_{ij}^D \delta_{ij}^D \bar{z}_{ij}^D \lambda_{ij}^D \right) - \sum_{l \in \bar{\mathcal{N}}} \psi_l \bar{\tau}_l \quad (28a)$$

$$\text{s.t. } \sum_{j \in \delta^+(i)} \left(x_{ij}^T \lambda_{ij}^T + x_{ij}^D \lambda_{ij}^D \right) - \sum_{j \in \delta^-(i)} \left(x_{ji}^T \lambda_{ji}^T + x_{ji}^D \lambda_{ji}^D \right) + \psi_i \geq 0, \quad \forall i \in \mathcal{N} \cup \{0'\}, \quad (28b)$$

$$\sum_{(i,j) \in \mathcal{A}} \left(\delta_{ij}^T + \delta_{ij}^D \right) \leq \Gamma, \quad (28c)$$

$$\lambda_{ij}^T \geq 0, \quad \forall (i, j) \in \mathcal{A}, \quad (28d)$$

$$\lambda_{ij}^D \geq 0, \quad \forall (i, j) \in \mathcal{A}, \quad (28e)$$

$$0 \leq \psi_l \leq 1, \quad \forall l \in \bar{\mathcal{N}}, \quad (28f)$$

$$\psi_l = 0, \quad \forall l \notin \overline{\mathcal{N}}, \quad (28g)$$

$$0 \leq \tilde{\lambda}_{ij}^T \leq \lambda_{ij}^T, \quad \forall (i, j) \in \mathcal{A}, \quad (28h)$$

$$0 \leq \tilde{\lambda}_{ij}^D \leq \lambda_{ij}^D, \quad \forall (i, j) \in \mathcal{A}, \quad (28i)$$

$$\lambda_{ij}^T - G(1 - \delta_{ij}^T) \leq \tilde{\lambda}_{ij}^T \leq G\delta_{ij}^T, \quad \forall (i, j) \in \mathcal{A}, \quad (28j)$$

$$\lambda_{ij}^D - G(1 - \delta_{ij}^D) \leq \tilde{\lambda}_{ij}^D \leq G\delta_{ij}^D, \quad \forall (i, j) \in \mathcal{A}, \quad (28k)$$

$$\delta_{ij}^T, \delta_{ij}^D \in \{0, 1\}, \quad \forall (i, j) \in \mathcal{A}, \quad (28l)$$

where $G > 0$ is a specified sufficient large number.

Proof. The problem (27) can be rewritten as

$$\max_{\delta, \lambda, \psi} \sum_{(i,j) \in \mathcal{A}} \left(x_{ij}^T z_{ij}^T \lambda_{ij}^T + x_{ij}^D z_{ij}^D \lambda_{ij}^D + x_{ij}^T \tilde{z}_{ij}^T \tilde{\lambda}_{ij}^T + x_{ij}^D \tilde{z}_{ij}^D \tilde{\lambda}_{ij}^D \right) - \sum_{l \in \overline{\mathcal{N}}} \psi_l \bar{\tau}_l \quad (29a)$$

$$\text{s.t.} \quad \sum_{j \in \delta^+(i)} \left(x_{ij}^T \lambda_{ij}^T + x_{ij}^D \lambda_{ij}^D \right) - \sum_{j \in \delta^-(i)} \left(x_{ji}^T \lambda_{ji}^T + x_{ji}^D \lambda_{ji}^D \right) + \psi_i \geq 0, \quad \forall i \in \mathcal{N} \cup \{0'\}, \quad (29b)$$

$$\sum_{(i,j) \in \mathcal{A}} \left(\delta_{ij}^T + \delta_{ij}^D \right) \leq \Gamma, \quad (29c)$$

$$\lambda_{ij}^T \geq 0, \quad \forall (i, j) \in \mathcal{A}, \quad (29d)$$

$$\lambda_{ij}^D \geq 0, \quad \forall (i, j) \in \mathcal{A}, \quad (29e)$$

$$0 \leq \psi_l \leq 1, \quad \forall l \in \overline{\mathcal{N}}, \quad (29f)$$

$$\psi_l = 0, \quad \forall l \notin \overline{\mathcal{N}}, \quad (29g)$$

$$\delta_{ij}^T, \delta_{ij}^D \in \{0, 1\}, \quad \forall (i, j) \in \mathcal{A}. \quad (29h)$$

As $\delta_{ij}^T, \delta_{ij}^D \in \{0, 1\}$, the bi-variable terms $\delta_{ij}^T \lambda_{ij}^T$ and $\delta_{ij}^D \lambda_{ij}^D$ can be replaced by new variables $\tilde{\lambda}_{ij}^T$ and $\tilde{\lambda}_{ij}^D$, respectively, which are constrained by $0 \leq \tilde{\lambda}_{ij}^T \leq \lambda_{ij}^T$, $0 \leq \tilde{\lambda}_{ij}^D \leq \lambda_{ij}^D$, $\lambda_{ij}^T - G(1 - \delta_{ij}^T) \leq \tilde{\lambda}_{ij}^T \leq G\delta_{ij}^T$, and $\lambda_{ij}^D - G(1 - \delta_{ij}^D) \leq \tilde{\lambda}_{ij}^D \leq G\delta_{ij}^D$, where $G > 0$ is a specified sufficient large number. By substituting the new variables into the previous bilinear optimization problem, we then derive the reformulation of problem (27), as presented in the Proposition. The assertion has been proved. $\square$

The number of binary variables in the above MILP problem is $2|\mathcal{A}|$, linearly depends on the size of the arcs, and thus the problem can be efficiently solved by commercial solvers in general.

6.3. Benders decomposition for the RTDRP-D

By the dual formulation presented in the previous subsection, we reformulate RTDRP-D as:

$$\min_{x \in \mathcal{X}} \max_{\delta, \lambda, \psi} \sum_{(i,j) \in \mathcal{A}} \left(x_{ij}^T z_{ij}^T \lambda_{ij}^T + x_{ij}^D z_{ij}^D \lambda_{ij}^D + x_{ij}^T \tilde{z}_{ij}^T \tilde{\lambda}_{ij}^T + x_{ij}^D \tilde{z}_{ij}^D \tilde{\lambda}_{ij}^D \right) - \sum_{l \in \overline{\mathcal{N}}} \psi_l \bar{\tau}_l, \quad (30)$$

which is NP-hard (Minoux, 2011) and difficult to be solved in general. As both Θ and $\Omega(x)$ are bounded polyhedrons, the optimal solutions of the second-stage MILP problem are extreme points of the polyhedrons. Though it is difficult to enumerate all the extreme points of the polyhedrons, we can develop a Benders decomposition (BD) algorithm to solve the main problem.

In each iteration of the BD algorithm, the MP is a binary program and provides a lower bound for the RTDRP-D, and the SP as presented in Proposition 6.1 is a MILP and yields an upper bound. The detailed procedures of the BD algorithm are presented in Algorithm 2 as follows.

In each iteration, the SP generates an optimality cut or a feasibility cut for the MP. Let $\mathcal{O}$ and $\mathcal{F}$ be the sets of optimal cuts and feasible cuts, respectively, then the MP can be formulated as:

$$\begin{aligned} \min_{\alpha, x} \quad & \alpha \\ \text{s.t.} \quad & \text{optimality cuts:} \\ & \alpha \geq \sum_{(i,j) \in \mathcal{A}} \left(x_{ij}^T z_{ij}^T \lambda_{ij}^T + x_{ij}^D z_{ij}^D \lambda_{ij}^D \right) - \sum_{l \in \overline{\mathcal{N}}} \psi_l \bar{\tau}_l, \quad \forall (z, \lambda, \psi) \in \mathcal{O}, \\ & \text{feasibility cuts:} \\ & \sum_{(i,j) \in \mathcal{A}} \left(x_{ij}^T z_{ij}^T \lambda_{ij}^T + x_{ij}^D z_{ij}^D \lambda_{ij}^D \right) - \sum_{l \in \overline{\mathcal{N}}} \psi_l \bar{\tau}_l \leq 0, \quad \forall (z, \lambda, \psi) \in \mathcal{F}, \\ & \alpha \text{ free, } x \in \mathcal{X}. \end{aligned} \quad (31)$$

As the sizes of the cut pools grow, the lower bound increases. The procedure stops when the gap between the bounds is reduced to be within an acceptable range.

Algorithm 2 Benders Decomposition for the RTDRP-D model.**Initialization:**

An feasible routing solution $\mathbf{x}_1 \in \mathcal{X}$ for iteration $m = 1$;
 Initialize the lower bound $L_{BD} = -\infty$ and the upper bound $U_{BD} = +\infty$;
 Specify a convergence tolerance level $\delta > 0$;
 Initialize the sets of feasibility and optimality cut pools $F = \mathcal{O} = \emptyset$.

Iterations:

For iteration $m \geq 1$,

1: Solve the subproblem (27) for $\mathbf{x} = \mathbf{x}_m$ by employing Proposition 6.1,

- if the subproblem is unbounded, denote by $(\delta_m, \tilde{\lambda}_m, \lambda_m, \psi_m)$ the unbounded rays, let $z_{m,ij}^T = \underline{z}_{ij}^T + \delta_{m,ij}^T \tilde{z}_{ij}^T$ and $z_{m,ij}^D = \underline{z}_{ij}^D + \delta_{m,ij}^D \tilde{z}_{ij}^D$ for any $(i, j) \in \mathcal{A}$. Add the feasibility cut

$$\sum_{(i,j) \in \mathcal{A}} \left(x_{ij}^T z_{m,ij}^T \lambda_{m,ij}^T + x_{ij}^D z_{m,ij}^D \lambda_{m,ij}^D \right) - \sum_{l \in \mathcal{N}^+} \psi_{m,l} \bar{r}_l \leq 0$$

to the cut pool, i.e., update $F := F \cup \{(\mathbf{z}_m, \lambda_m, \psi_m)\}$.

- if the subproblem is bounded, denote by $\rho(\mathbf{x}_m)$ the optimal objective value and $(\delta_m, \tilde{\lambda}_m, \lambda_m, \psi_m)$ the optimum. Update $U_{BD} = \min \{U_{BD}, \rho(\mathbf{x}_m)\}$. Let $z_{m,ij}^T = \underline{z}_{ij}^T + \delta_{m,ij}^T \tilde{z}_{ij}^T$ and $z_{m,ij}^D = \underline{z}_{ij}^D + \delta_{m,ij}^D \tilde{z}_{ij}^D$ for any $(i, j) \in \mathcal{A}$, add the optimality cut

$$\alpha \geq \sum_{(i,j) \in \mathcal{A}} \left(x_{ij}^T z_{m,ij}^T \lambda_{m,ij}^T + x_{ij}^D z_{m,ij}^D \lambda_{m,ij}^D \right) - \sum_{l \in \mathcal{N}^+} \psi_{m,l} \bar{r}_l$$

to the cut pool, i.e., update $\mathcal{O} := \mathcal{O} \cup \{(\mathbf{z}_m, \lambda_m, \psi_m)\}$.

- 2: Check the convergence level. If $U_{BD} - L_{BD} < \delta$, then stop the algorithm and return $\mathbf{x}_m$ as the optimal routing solution of the RTDRP-D. Otherwise, go to Step 3.
- 3: Solve the master problem (31) with the updated cut pool. Let $(\mathbf{x}_m, \alpha_m)$ be the optimal solution, update $L_{BD} = \alpha_m$. Update $m := m + 1$, and go back to Step 1.

6.4. Valid inequalities

In this subsection, we develop several valid inequalities to enhance the computational efficiency of the proposed BD algorithms. At first, we introduce

$$t_{0'} \geq \sum_{(i,j) \in \mathcal{A}} \left(s_i^T + z_{ij}^T \right) x_{ij}^T, \quad (32)$$

as a lower bound on the total tour time. The inequality (32) states that the arrival time of the truck at the end depot is no less than the total time required by the truck to travel all the arcs covered by $\mathbf{x}$. Similarly,

$$t_{0'} \geq \sum_{(i,j) \in \mathcal{A}} \left(s_i^D + z_{ij}^D \right) x_{ij}^D, \quad (33)$$

imposes that the total tour duration is no less than the sum of the travel times of the arcs traversed by the drone. Although the inequalities (32)–(33) seem trivial, they can greatly accelerate the computation process (Roberti and Ruthmair, 2021).

In addition to the above inequalities based on the total tour duration, we introduce new upper bounds on service starting time of nodes visited by the truck or the drone alone. Suppose a node j is traversed by the truck alone, then the optimal service starting time t_j^* should be equal to $t_i + s_i^T + z_{ij}^T$, where i is the previous node traversed by the truck before node j . Thus, the feasible route set can be strengthened by the following big-M constraint:

$$t_j \leq \sum_{i \in \delta^-(j)} \left(t_i + s_i^T + z_{ij}^T \right) x_{ij}^T + M \left(1 - y_j^T \right), \quad \forall j \in \mathcal{N}, \quad (34)$$

where M is a sufficient large positive number. By the same token, we can introduce the inequalities for service starting time of nodes traversed by the drone alone:

$$t_j \leq \sum_{i \in \delta^-(j)} \left(t_i + s_i^D + z_{ij}^D \right) x_{ij}^D + M \left(1 - y_j^D \right), \quad \forall j \in \mathcal{N}. \quad (35)$$

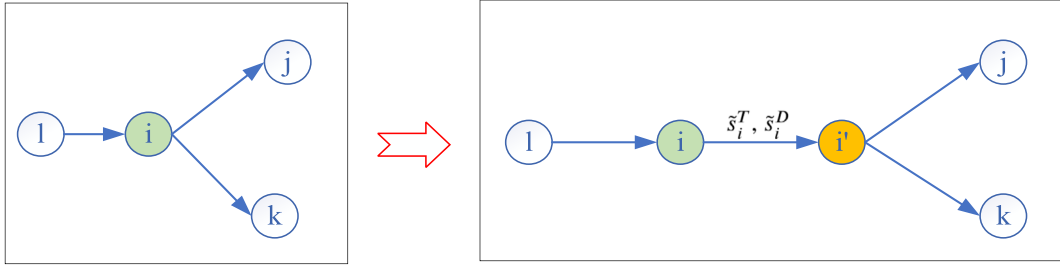


Fig. 5. The extended network with random service time.

One can see that, the inequalities (34)–(35) cover the majority of the nodes, except for the pairs $(i, j) \in \mathcal{A}$ with $y_i^C = y_j^C = 1$ and $x_{ij}^T = 1$. By Assumption (b), if node j is traversed directly after node i by the truck in tandem with the drone, the optimal service arrival time of node j is $t_j^* = t_i + s_i^T + z_{ij}^T$. This motivates us to introduce the following inequality:

$$t_j \leq t_i + s_i^T + z_{ij}^T + M(3 - x_{ij}^T - y_i^C - y_j^C), \quad \forall j \in \mathcal{N}, i \in \delta^-(j). \quad (36)$$

When $y_i^C = y_j^C = 1$ and $x_{ij}^T = 1$, the service starting time t_j is upper bounded by $t_i + s_i^T + z_{ij}^T$; otherwise, it is upper bounded by M . All nodes in $\mathcal{N}$ are involved in one of the inequalities (34)–(36), which provide efficient cuts to improve the general solution process. Per our knowledge, the previous three inequalities are the first to appear in literature on truck-and-drone routing problems.

7. Extensions

7.1. The uncertainty of service time

Service time uncertainty is another factor worth considering in truck-and-drone routing problems. However, travel time uncertainty is deemed the most critical factor in this study, because the uncertainty of service time can be reading incorporated into the proposed STDPR-D and RTDRP-D models.

When considering the uncertainty of service time, we can add a dummy node for each node, as shown in Fig. 5. Specifically, for each $i \in \mathcal{N}$, we add a dummy node i' and an arc (i, i') in the complete directed graph G ; furthermore, we replace each arc (i, j) with $j \in \delta^+(i)$ by a new arc (i', j) . The random travel time of arc (i, i') can be imposed the same as the service time of node i , i.e., $\bar{z}_{ii'}^T = \bar{s}_i^T$, $\bar{z}_{ii'}^D = \bar{s}_i^D$. Moreover, for each $j \in \delta^+(i)$, we impose that $\bar{z}_{i'j}^T = \bar{z}_{ij}^T$, $\bar{z}_{i'j}^D = \bar{z}_{ij}^D$. Then, the proposed STDPR-D and RTDRP-D models, as well as the BD algorithms, can be generalized by jointly considering the uncertainties of travel time and service time.

7.2. Drone payload and flying range

If the drone has a maximum payload Q^D , that is, the drone cannot delivery a package with weight higher than Q^D , then we can add the following constraint to set formulation (1):

$$q_i \leq Q^D + M(1 - y_i^D), \quad \forall i \in \mathcal{N}, \quad (37)$$

where q_i is the package weight of customer i , M be a sufficient large number. The constraint guarantees that if the weight of customer i 's package exceeds the drone payload, the customer cannot be serviced by the drone alone.

The payload and flying range of the drone depend on its remaining battery energy. Assume the drone has a limited battery capacity B^D . The energy consumption is closely related to the package weight and the distance of the arc. For each arc $(i, j) \in \mathcal{A}$, one can precompute the energy consumption value $b_{ij}^\oplus$ of traversing the arc (i, j) with the package of customer j is on board, as well as the energy consumption value b_{ij}° of traversing the arc (i, j) without any package. Under this assumption, we can take the energy-dependent flying range into account:

$$\sum_{(j,i) \in \mathcal{A}} b_{ji}^\oplus x_{ji}^D + \sum_{(i,k) \in \mathcal{A}} b_{ik}^\circ x_{ik}^D \leq B^D + M(1 - y_i^D), \quad \forall i \in \mathcal{N}. \quad (38)$$

The constraint ensures that if the drone visits a customer i alone, i.e., $y_i^D = 1$, $i \in \mathcal{N}$, the total energy consumption values of the arcs traversed by the drone and incident to i does not exceed the battery energy capacity.

7.3. Soft time windows

The restriction of deadlines can be extended to the scenario with soft time windows. In that case, each node $i \in \mathcal{N}$ is associated with a time window $[\underline{\tau}_i, \bar{\tau}_i]$, where $\underline{\tau}_i \in \mathbb{R}_+$ and $\bar{\tau}_i \in \mathbb{R}_+ \cup \{+\infty\}$ represent the earliest and latest times for providing delivery services

to customer i . It is possible that some customers do not specify time windows. In such cases, we set $\underline{\tau}_i = 0$ when no earliest time is specified for customer i and $\bar{\tau}_i = +\infty$ when no latest time is specified. Denote by

$$\underline{\mathcal{N}} := \{i \in \mathcal{N} \mid \underline{\tau}_i > 0\}, \quad \overline{\mathcal{N}} := \{i \in \mathcal{N} \mid \bar{\tau}_i < +\infty\},$$

the sets of customers with positive earliest times and finite latest times, respectively.

It is reasonable to assume that, the drone is not allowed to hover over customers and wait for the earliest service time to arrive before providing service to customers. This is typically the case for truck-and-drone delivery. In this sense, time window is “soft” indicates that the service may start before $\underline{\tau}_i$ or after the deadline $\bar{\tau}_i$, but both will lead to penalties.

STDRP with soft time windows The min-sum STDRP model with soft time windows can be formulated as

$$\min_{\mathbf{x} \in \mathcal{X}} \mathbb{E}_{\mathbb{P}} \left\{ \sum_{l \in \underline{\mathcal{N}}} \left[(\underline{\tau}_l - t_l(\mathbf{x}, \mathbf{z}))^+ \right] + \sum_{l \in \overline{\mathcal{N}}} \left[(t_l(\mathbf{x}, \mathbf{z}) - \bar{\tau}_l)^+ \right] \right\}. \quad (39)$$

As an extension of (8), it can be approximated by the following LP:

$$\min_{\mathbf{x}, \{\xi_l^k, \eta_l^k, t_{ij}^k\}_{k=1, \dots, K}} \frac{1}{K} \sum_{k=1}^K \left(\sum_{l \in \underline{\mathcal{N}}} \xi_l^k + \sum_{l \in \overline{\mathcal{N}}} \eta_l^k \right) \quad (40a)$$

$$\text{s.t.} \quad t_i^k + s_i^T + z_{ij}^T \leq t_j^k + M(1 - x_{ij}^T), \quad \forall (i, j) \in \mathcal{A}, \forall k = 1, \dots, K, \quad (40b)$$

$$t_i^k + s_i^D + z_{ij}^D \leq t_j^k + M(1 - x_{ij}^D), \quad \forall (i, j) \in \mathcal{A}, \forall k = 1, \dots, K, \quad (40c)$$

$$\xi_l^k \geq \underline{\tau}_l - t_l^k, \quad \forall l \in \underline{\mathcal{N}}, \forall k = 1, \dots, K, \quad (40d)$$

$$\eta_l^k \geq t_l^k - \bar{\tau}_l, \quad \forall l \in \overline{\mathcal{N}}, \forall k = 1, \dots, K, \quad (40e)$$

$$\xi_l^k \geq 0, \quad \forall l \in \underline{\mathcal{N}}, \forall k = 1, \dots, K, \quad (40f)$$

$$\eta_l^k \geq 0, \quad \forall l \in \overline{\mathcal{N}}, \forall k = 1, \dots, K, \quad (40g)$$

$$\mathbf{x} \in \mathcal{X}. \quad (40h)$$

The min-max STDRP model with soft time windows can be established in a similar way.

RTDRP with soft time windows The RTDPR model with soft time windows, as an extension of (14), can be stated as follows:

$$\min_{\mathbf{x} \in \mathcal{X}} \max_{\mathbf{z} \in \mathcal{Z}} \min_{\xi, \eta, t} \sum_{l \in \underline{\mathcal{N}}} \xi_l + \sum_{l \in \overline{\mathcal{N}}} \eta_l \quad (41a)$$

$$\text{s.t.} \quad t_i + s_i^T + z_{ij}^T \leq t_j + M(1 - x_{ij}^T), \quad \forall (i, j) \in \mathcal{A}, \quad (41b)$$

$$t_i + s_i^D + z_{ij}^D \leq t_j + M(1 - x_{ij}^D), \quad \forall (i, j) \in \mathcal{A}, \quad (41c)$$

$$\xi_l \geq \underline{\tau}_l - t_l, \quad \forall l \in \underline{\mathcal{N}}, \quad (41d)$$

$$\eta_l \geq t_l - \bar{\tau}_l, \quad \forall l \in \overline{\mathcal{N}}, \quad (41e)$$

$$\xi_l \geq 0, \quad \forall l \in \underline{\mathcal{N}}, \quad (41f)$$

$$\eta_l \geq 0, \quad \forall l \in \overline{\mathcal{N}}. \quad (41g)$$

The solving procedures of the STDRP and RTDRP models with soft time windows closely mirror those used in solving STDRP-D and RTDRP-D. The details of the solving procedure are omitted here for brevity.

7.4. Nonlinear objective functions

In this subsection, we extend the analysis to the STDRP-D problems with nonlinear objective functions. Specifically, we take $\rho(\mathbf{x}) = \sum_{l \in \overline{\mathcal{N}}} \mathbb{P}(t_l(\mathbf{x}, \mathbf{z}) - \bar{\tau}_l)$ as the service delay index, where $\mathbb{P}$ is a given probability distribution of uncertain travel times. One wishes to minimize the sum of probability of service delays under the distribution $\mathbb{P}$, i.e., she solves

$$\min_{\mathbf{x}} \sum_{l \in \overline{\mathcal{N}}} \mathbb{P}(t_l(\mathbf{x}, \mathbf{z}) - \bar{\tau}_l). \quad (42)$$

The problem is nonlinear and generally difficult to solve.

The proposed SAA approach and its associated BD algorithm can also be employed to overcome the computational difficulty. Denote by $\{\hat{z}_1, \hat{z}_2, \dots, \hat{z}_K\}$ the samples of travel time. We introduce a variable $\phi_l^k \in \{0, 1\}$ to indicate whether the deadline at node $l \in \overline{\mathcal{N}}$ is violated in the sample scenario k . If the service start time of node l in scenario k exceeds the deadline, ϕ_l^k is equal to 1, otherwise it equals 0. The nonlinear probability function $\rho(\mathbf{x}) = \sum_{l \in \overline{\mathcal{N}}} \mathbb{P}(t_l(\mathbf{x}, \mathbf{z}) - \bar{\tau}_l)$ can be approximated by the expected number

Table 3
Initial parameters setting for numerical experiments.

Parameters	Description	Values	Reference
$(N, \mathcal{A})$	Numbers of customers and arcs	(10,30),(15,40),(20,60),(25,90)	Poikonen et al. (2019)
(x_i, y_i)	Position coordinates of customers i	$\sim U(0, 50) \times U(0, 50)$	Poikonen et al. (2019)
K	Sample size	100,200,500	–
θ	Relative speed of the drone to that of the truck	1.5	Roberti and Ruthmair (2021)
$\underline{z}_{ij}^T$	Nominal truck travel time of arc (i, j)	$\lfloor x_i - x_j + y_i - y_j \rfloor$	Roberti and Ruthmair (2021)
$\underline{z}_{ij}^D$	Nominal drone travel time of arc (i, j)	$\lfloor \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} / \theta \rfloor$	Roberti and Ruthmair (2021)
$\hat{z}_{ij}^T$	Maximum variation of truck travel time over arc (i, j)	$0.5 \underline{z}_{ij}^T$	–
$\hat{z}_{ij}^D$	Maximum variation of drone travel time over arc (i, j)	$0.5 \underline{z}_{ij}^D$	–
Γ	Budget of uncertain set	$\left[\frac{1}{K} \sum_{k=1}^K \sum_{a \in \mathcal{A}} \left[\frac{\hat{z}_{i,a}^T - \underline{z}_{i,a}^T}{\underline{z}_{i,a}^T} + \frac{\hat{z}_{i,a}^D - \underline{z}_{i,a}^D}{\underline{z}_{i,a}^D} \right] \right]$	–
s_i^T	Truck service time of customer i	$\sim U(1, 5)$	–
s_i^D	Drone service time of customer i	$\sim U(1, 5)$	–
$\bar{e}_i$	Service deadline of customer i	$[0.5 \hat{t}_{0'}, \hat{t}_{0'}]$	–

of deadline violations, i.e., $\frac{1}{K} \sum_{k=1}^K \sum_{l \in \mathcal{N}} \phi_l^k$. Then, the nonlinear problem (42) can be reformulated as follows:

$$\min_{\mathbf{x}, \{\phi_l^k\}_{k=1, \dots, K}} \frac{1}{K} \sum_{k=1}^K \sum_{l \in \mathcal{N}} \phi_l^k \quad (43a)$$

$$\text{s.t.} \quad t_i^k + s_i^T + \hat{z}_{k,ij}^T \leq t_j^k + M(1 - x_{ij}^T), \quad \forall (i, j) \in \mathcal{A}, \forall k = 1, \dots, K, \quad (43b)$$

$$t_i^k + s_i^D + \hat{z}_{k,ij}^D \leq t_j^k + M(1 - x_{ij}^D), \quad \forall (i, j) \in \mathcal{A}, \forall k = 1, \dots, K, \quad (43c)$$

$$t_l^k \leq \bar{e}_l + M\phi_l^k, \quad \forall l \in \mathcal{N}, \forall k = 1, \dots, K, \quad (43d)$$

$$\phi_l^k \in \{0, 1\}, \quad \forall l \in \mathcal{N}, \forall k = 1, \dots, K, \quad (43e)$$

$$\mathbf{x} \in \mathcal{X}, \quad (43f)$$

where M is a specified sufficient large positive number. The third constraint $t_l^k \leq \bar{e}_l + M\phi_l^k$ enforces ϕ_l^k to be 1 if the deadline at node l is violated with respect to the service start time under scenario k , i.e., $t_l^k > \bar{e}_l$.

The challenge in solving problem (43) is due to the numbers of binary decision variables ϕ_l^k and the third constraints grow as the sample size increases. Nevertheless, a BD method similar to Algorithm 1 can be developed to efficiently solve the problem, because it has a similar structure to problem (8). To void duplication, we omit the detailed solving procedure here.

8. Numerical experiments

We conduct numerical experiments to test the performance of the proposed model and the BD algorithm. The test instances are generated in a similar way to those employed in Poikonen et al. (2019) and Roberti and Ruthmair (2021). Specifically, we consider instances with N customers (e.g., $N = 10, 15, 20$, or 25) located on a 50×50 grid. The coordinates of the depot and all the customers are randomly generated following a uniform distribution.

For any two nodes i and j with coordinates (x_i, y_i) and (x_j, y_j) , we use the budget uncertain set (10) to model the uncertainty of travel time over the arc (i, j) . Following the expressions of travel time formulated in Roberti and Ruthmair (2021), we set the nominal truck travel time $\underline{z}_{ij}^T$ and the nominal drone travel time $\underline{z}_{ij}^D$ as $\underline{z}_{ij}^T = \lfloor |x_i - x_j| + |y_i - y_j| \rfloor$ and $\underline{z}_{ij}^D = \lfloor \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} / \theta \rfloor$, respectively, both depends on the distance between node i and j . The parameter $\theta > 0$ is used to characterize the relative speed of the drone to that of the truck. For each arc $(i, j) \in \mathcal{A}$, We set the variation ranges of the travel time to be $\hat{z}_{ij}^T = 0.5 \underline{z}_{ij}^T$ and $\hat{z}_{ij}^D = 0.5 \underline{z}_{ij}^D$.

For each arc $(i, j) \in \mathcal{A}$, we generate K samples of travel time $\hat{z}_{k,ij}^T$ and $\hat{z}_{k,ij}^D$, $k = 1, \dots, K$. The samples $\hat{z}_{k,ij}^T$ and $\hat{z}_{k,ij}^D$ are randomly generated in the ranges $[\underline{z}_{ij}^T, \underline{z}_{ij}^T + \hat{z}_{ij}^T]$ and $[\underline{z}_{ij}^D, \underline{z}_{ij}^D + \hat{z}_{ij}^D]$, respectively, following the uniform distribution. We estimate the uncertainty budget Γ from the samples by (12). Based on the samples $\{\hat{z}_1, \hat{z}_2, \dots, \hat{z}_K\}$, we derive the optimal total tour duration $\hat{t}_{0'}$ to server all customers by solving the SAA model (7). For each customer $i \in \mathcal{N}$, the deadline $\bar{e}_i$ is randomly selected in the range $[0.5 \hat{t}_{0'}, \hat{t}_{0'}]$, both the service time s_i^T, s_i^D are randomly generated in the ranges $[1, 5]$ following a uniform distribution. We summarize all the parameter settings in Table 3.

To verify the performance of the proposed BD algorithms, we conduct algorithmic comparison experiments among the following approaches:

- **BD**: The Benders decomposition algorithm established in this study while no valid inequalities is employed. It refers to Algorithms 1 and 2 in the stochastic and robust scenarios, respectively.
- **BD⁺**: The Benders decomposition algorithms with valid inequalities (32)–(36).
- **BC**: The branch-and-cut approach for solving the stochastic routing problem with deadlines.

Table 4
Average results on the STDRP-D instances with deadlines.

N	A	K	Prob (%)	Num	Risk	TT	CPU			Gap	
							BC	BD	BD ⁺	G_{BC}	G_{BD}
10	30	100	10.4	2.99	19.78	342.98	89.56	81.06	73.24	18.22	9.65
		200	5.60	1.27	8.36	353.41	92.41	92.75	75.26	18.56	18.86
		500	3.82	0.85	6.42	349.56	109.26	106.43	81.43	25.47	23.49
15	40	100	11.02	3.93	20.74	421.05	204.57	192.45	172.48	15.69	10.38
		200	8.05	2.04	8.47	426.30	222.72	216.38	189.56	14.89	12.39
		500	4.24	1.39	6.68	431.02	265.78	245.17	194.78	26.71	20.55
20	60	100	20.78	6.90	30.36	623.90	668.72	634.25	567.49	15.14	10.53
		200	9.21	2.84	12.45	681.10	842.28	782.17	638.25	24.22	18.40
		500	6.05	1.46	9.62	573.30	876.48	804.36	647.34	26.14	19.52
25	90	100	20.85	7.87	40.01	770.87	1082.12	1034.72	902.91	16.56	12.74
		200	12.01	3.15	16.48	981.75	1478.24	1307.39	1021.18	30.92	21.89
		500	8.46	1.82	13.05	802.37	1607.28	1536.37	1128.56	29.78	26.54

- **OA**: The outer approximation algorithm for solving two-stage adaptive robust optimization problems. We use the OA approach developed by Bertsimas et al. (2013) to solve the RTDRP-D and compare its performance with the BD algorithm proposed in this study.

The algorithms are coded in Python 3.6.5 under Windows 10 and solved by the GUROBI 9.1.1. All the experiments are performed on a laptop with a 4-core 3.20 GHz CPU and 16 GB RAM.

8.1. Performance comparison on the STDRP-D

In this subsection, we compare the performance of the BC, BD, BD⁺ on the stochastic truck-and-drone routing problem. Kenyon and Morton (2003) exploited a branch-and-cut (BC) approach to solve stochastic vehicle routing problems with deadlines. We use this BC approach as a benchmark to solve the STDRP-D. To test the efficiency of the valid inequalities (32)–(36), we compare the performance of the BD Algorithm 1 and its corresponding BD⁺ algorithm.

We consider four scenarios of customer network size, $(N, |A|) = (10, 30), (15, 40), (20, 60), (25, 90)$, and vary the size of samples K in $\{100, 200, 500\}$ to examine its impact on the solution performance and computational efficiency. For each scenario of parameters setting, 200 instances are generated. To assess the out-of-sample performance of the models, an additional 10,000 independent samples of $\mathbf{z}$ are generated. For each testing instance, we solve the three models and derive their optimal solutions, then compute the performance indicators of each solution.

After testing all 200 instances, we calculate the mean values of the performance indicators obtained in each instance. Specifically, the following indicators are used:

- **Prob**: the probability that service lateness occurs, i.e., $\mathbb{P}(\rho(\mathbf{x}, \mathbf{z}) > 0)$;
- **Num**: the mean number of customers whose service is delayed when service lateness occurs, i.e.,

$$\mathbb{E} \left[\# \left(l \in \overline{\mathcal{N}} : \xi_l(\mathbf{x}, \mathbf{z}) > 0 \right) \mid \rho(\mathbf{x}, \mathbf{z}) > 0 \right];$$

- **Risk**: the mean service delay risk index, i.e., $\mathbb{E}[\rho(\mathbf{x}, \mathbf{z}) \mid \rho(\mathbf{x}, \mathbf{z}) > 0]$;
- **TT**: the mean total travel time, i.e., $\mathbb{E}[t_{0r}(\mathbf{x}, \mathbf{z})]$;
- **CPU**: the mean running time (in seconds);
- **Gap**: the computational efficiency improvement rate of BD⁺, computed as $G_{Algo} = \frac{CT_{Algo} - CT_{BD^+}}{CT_{Algo}} \times 100\%$, where $Algo \in \{BC, BD, OA\}$, and CT represents the CPU Time consumed by the algorithms.

In the definitions, the probability $\mathbb{P}$ and the expectation $\mathbb{E}$ are computed over the 10,000 out-of-sample data points.

We evaluate the performance of the algorithms on all the scenarios of parameters setting and show the results in Table 4. Note that the performance of the three approaches is very close to each other in terms of the indicators including *Prob*, *Num* and *Risk*, with the main difference being in computational efficiency. One can see that, the indicator values *Prob*, *Num*, *Risk*, are all decreasing as the sample size K increases. This is intuitive because as the sample size increases, we can get access to more information on travel time distribution, and the impact of uncertainty will decrease, then the performance of the routing solution will be better.

As shown in Table 4, both BD and BD⁺ outperforms BC in terms of CPU Time. All of the instances are solved to optimality by the BD⁺ in the lowest seconds. The BD⁺ is also more efficient than the BD, which demonstrates the efficiency of the proposed valid inequalities. We further note that BD⁺ is less sensitive to the size of the problem, as both the gaps G_{BC} and G_{BD} generally increases when $(N, |A|, K)$ grows. Therefore, the BD⁺ algorithm is far superior to the other two algorithms in term of CPU Time especially when the size of the problem is large. Although its performance is not yet perfect, as the computational effort increases exponentially as the scale size increases linearly (e.g. from 10 nodes to 20 nodes within 100 sample sizes), it paves the way for further improvement.

Table 5
Average results on the RTDRP-D instances with deadlines.

N	A	Prob (%)	Num	Risk	TT	CPU			Gap	
						OA	BD	BD ⁺	G_{OA}	G_{BD}
10	30	12.9	2.88	21.08	368.48	134.36	89.78	75.42	43.87	15.99
15	40	13.5	3.25	23.28	417.84	316.48	204.50	165.38	47.74	19.13
20	60	18.7	5.91	27.92	615.79	1578.36	942.06	762.62	51.68	19.05
25	90	19.3	7.16	34.61	834.62	4172.66	2543.36	1973.43	52.71	22.41

Finally, it is important to note that the proposed algorithm provides an exact routing solution, which is not necessary in practice, whereas a near-optimal solution is enough. Therefore, we can develop heuristic algorithms based on this method to reduce computational time and improve practicability.

8.2. Performance comparison on the RTDRP-D

The RTDRP-D model (13) is actually a two-stage adaptive robust optimization problem, which is difficult to solve in general. Bertsimas et al. (2013) developed a combination of Benders decomposition type algorithm and an outer approximation (OA) technique to solve it. A common difficulty encountered in solving two-stage adaptive robust optimization problems is how to effectively solve bilinear subproblems, which is the motivation behind the proposal of the OA algorithm in Bertsimas et al. (2013). However, we can transform the bilinear problem (27) into a MILP problem in this study, this enables us to propose the exact BD Algorithm 2 to solve the RTDRP-D.

We perform experiments in this subsection to compare the performance of the OA, BD and BD⁺ algorithms. We test the algorithms on four scenarios $(N, |A|) = (10, 30), (15, 40), (20, 60), (25, 90)$, as used in the previous subsection. And we set the sample size $K = 100$ and do not vary its value, as the sample data is only used to estimate the parameters of the uncertain set and its size has little impact on the computational efficiency of the algorithms. The results are shown in Table 5. One can observe that, the risk of deadline violation is increasing, as the size of the problem grows. All the results in terms of *Prob*, *Num* and *Risk* are supporting this observation.

In terms of computational efficiency, the average CPU times of both the BD and BD⁺ are significantly lower than that of the OA. This validates the value of Proposition 6.1, which transforms the bilinear problem (27) into a MILP instead of solving it through heuristic methods such as the OA, and thus greatly improving computational efficiency. Furthermore, the results on columns *Gap* indicate that, the valid inequalities (32)–(36) can greatly boost the computational efficiency. We conclude that, the proposed BD algorithm together with the valid inequalities provide an efficient approach for solving the two-stage RTDRP-D model.

8.3. Robustness against travel time distribution

In practice, one usually has no access to the true distribution of travel times. Therefore, it is possible that the samples are generated from a mis-specified distribution. In this subsection, we test the robustness of the proposed model. For a comparison, we adopt the following three approaches:

- DM: Solving the deterministic TDRP-D problem (21), in which the uncertain travel time $z_{ij}^T, z_{ij}^D ((i, j) \in A)$ are replaced by their sample means $\bar{z}_{ij}^T := \frac{1}{K} \sum_{k=1}^K \hat{z}_{k,ij}^T$ and $\bar{z}_{ij}^D := \frac{1}{K} \sum_{k=1}^K \hat{z}_{k,ij}^D$, respectively.
- SAA: Solve the STDRP-D (7) by the BD Algorithm 1.
- RO: Solving the RTDRP-D problem (14) by the BD Algorithm 2.

We test their performance under scenarios with different true distributions of travel times. Specifically, the sample data are still generated from a uniform distribution, but the testing data are generated from the true distribution. We consider four scenarios of the true distribution realization: (i) the same uniform distribution used for sample data generation; (ii) the normal distribution whose mean and variance are estimated from the sample data; (iii) Beta(2, 4) distribution; and (iv) Beta(4, 2) distribution.

We generate the sample data with size $K = 100$ by the same procedure as employed in the previous subsection. Subsequently, we formulate the corresponding DM, SAA, and RO models of TDRP-D, finally deriving the routing solutions for each model. For each scenario of true distribution realization, we generate 10,000 instances of testing data following the true distribution. We use the indicators *Prob*, *Num*, *Risk*, *TT* to measure the models' robustness under different true distributions. All testing results are reported in Table 6.

We observe that the RO model outperforms the other benchmarks in all scenarios. This observation is consistent with our expectations. When the distribution is misspecified, the statistical information estimated from the samples may deviate significantly from its true value. Building upon the mis-estimated information, the routing solutions derived by the DM and SAA models have a high probability of poor performance under the true distribution. In contrast, the RO model does not overly rely on the information estimated from sample data, but considers all possible realizations of travel time in the model, obtaining a routing solution whose worst-case value is optimal.

Table 6

Robustness of the routing solutions derived by the models under different true distributions.

True distribution	Model	Prob (%)	Num	Risk	TT
Uniform	DM	32.9	5.78	27.26	337.88
	SAA	10.4	2.99	19.78	342.98
	RO	8.70	2.26	17.43	355.43
Normal	DM	48.76	7.16	40.39	341.27
	SAA	18.97	3.23	28.42	312.23
	RO	13.25	2.15	22.21	315.07
Beta(2,4)	DM	59.06	6.37	42.41	340.78
	SAA	22.43	3.61	31.55	361.10
	RO	15.21	2.34	24.16	314.72
Beta(4,2)	DM	62.10	5.82	63.59	354.12
	SAA	28.46	3.15	50.14	364.47
	RO	21.53	2.23	31.78	311.03

Table 7

Average computational efficiency of the BD algorithm for the STDPR-D.

N	A	K	Numbers of		Run time (s) of		
			Iterations	Optimality cuts	MP	SP	BD
10	30	100	22.90	7.9	36.71	43.47	81.06
		200	24.12	8.1	37.28	56.42	92.75
		500	25.31	8.8	39.06	68.76	106.43
15	40	100	57.13	15.2	79.20	115.17	192.45
		200	59.26	15.8	91.13	122.98	216.38
		500	61.33	16.1	105.32	140.02	245.17
20	60	100	153.24	23.1	281.36	350.77	634.25
		200	161.37	26.2	292.09	489.15	782.17
		500	170.23	26.9	317.38	488.21	804.36
25	90	100	202.17	30.4	426.39	609.13	1034.72
		200	215.43	32.1	451.76	849.34	1307.39
		500	232.25	35.8	485.32	1050.92	1536.37

Table 8

Average computational efficiency of the BD algorithm for the RTDRP-D.

N	A	Numbers of		Run time (s) of		
		Iterations	Optimality cuts	MP	SP	BD
10	30	24.2	8.3	43.41	45.36	89.78
15	40	58.3	14.5	112.33	94.16	204.50
20	60	182.6	36.1	516.28	435.78	942.06
25	90	361.9	43.2	1423.06	1109.78	2543.36

8.4. Iteration and cut numbers of the BD

In this subsection, we examined the computational efficiency of the BD algorithm under different sizes of the system. We consider four scenarios: (i) $N = 10$, $|A| = 30$; (ii) $N = 15$, $|A| = 40$; (iii) $N = 20$, $|A| = 60$; and (iv) $N = 25$, $|A| = 90$. For each scenario, the sample size was varied among 100, 200, 500 to assess its impact on computational efficiency. We present the testing results for the BD Algorithms 1 and 2 in Tables 7 and 8, respectively. For each instance, we report the numbers of the iterations, the optimality cuts used by the algorithms, as well as the running times respectively consumed by the master problem (MP), the subproblem (SP) and the whole algorithm. For each size of the problem, we conduct the experiments for 20 independent instances, and evaluate the means of the indicator values among the instances.

We observe from the results that, the running time significantly increases with the size of the problem, while being less affected by the sample size. The numbers of optimality cuts and iterations increase with the size of the problem, but the numbers of feasible cuts do not show a substantial increase. The computational efficiency is greatly affected by the choice of the initial routing solution. The computational efficiency is noticeably influenced by the choice of the initial routing solution, emphasizing the importance of an appropriately initialized routing solution. The results demonstrate that the BD algorithms proposed in this study can significantly accelerate the running time for solving large-scale TDRP-D problems without sacrificing performance.

8.5. Benefits of truck-and-drone tandems

To get insights into the benefits of truck-and-drone tandems, we compare the performance of a truck-and-drone delivery system with that of traditional vehicle delivery system. For stochastic routing problems, we consider the same scenarios of parameters

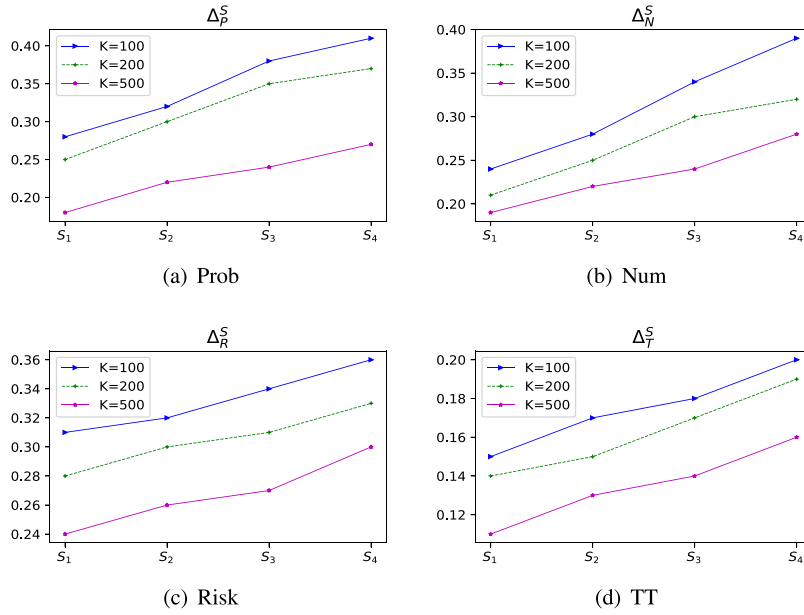


Fig. 6. Benefits of truck-and-drone tandems in stochastic routing problems.

setting as Section 8.1, i.e., $(N, |A|) = (10, 30), (15, 40), (20, 60), (25, 90)$ and $K \in \{100, 200, 500\}$. For simplicity, we denote the four scenarios of the value setting of $(N, |A|)$ as S_1, S_2, S_3 and S_4 , respectively. For instance, in scenario S_1 , $(N, |A|) = (10, 30)$.

For each scenario of parameters setting, we generate 200 instances of the samples with size K . For each instance of samples, we first solve the corresponding STDRP-D model (8) and derive its optimal routing solution $\mathbf{x}_{\text{STDRP}}^*$. Then, we add a constraint $y_i^D = 0, \forall i \in \mathcal{N}$ into the feasible set (1), and find an optimal routing solution to the STDRP-D model (8) in the modified feasible set, the problem is equivalent to a traditional stochastic vehicle routing problem with deadlines (Adulyasak and Jaillet, 2016). Hence, we may denote its optimal solution by $\mathbf{x}_{\text{SVRP}}^*$. We test the out-of-sample performance of the routing solutions on another 10,000 independent samples of $\mathbf{z}$, and calculate their performance index values of *Prob*, *Num*, *Risk*, *TT* (as introduced in Section 8.1).

We repeat the above procedure for all the 200 instances, and finally get 200 instances of the performance index values for both the solutions $\mathbf{x}_{\text{STDRP}}^*$ and $\mathbf{x}_{\text{SVRP}}^*$. Denote by $\text{Prob}_{\text{STDRP}}^S$, $\text{Num}_{\text{STDRP}}^S$, $\text{Risk}_{\text{STDRP}}^S$, $\text{TT}_{\text{STDRP}}^S$ the mean performance index values of $\mathbf{x}_{\text{STDRP}}^*$. Similarly, $\text{Prob}_{\text{SVRP}}^S$, $\text{Num}_{\text{SVRP}}^S$, $\text{Risk}_{\text{SVRP}}^S$, $\text{TT}_{\text{SVRP}}^S$ represent the mean performance index values of $\mathbf{x}_{\text{SVRP}}^*$. To measure the benefits of truck-and-drone tandems for stochastic routing problems, we define the reduction rates of deadline violation indices as follows:

$$\Delta_P^S := \frac{\text{Prob}_{\text{SVRP}}^S - \text{Prob}_{\text{STDRP}}^S}{\text{Prob}_{\text{SVRP}}^S}, \quad \Delta_N^S := \frac{\text{Num}_{\text{SVRP}}^S - \text{Num}_{\text{STDRP}}^S}{\text{Num}_{\text{SVRP}}^S}, \quad (44a)$$

$$\Delta_R^S := \frac{\text{Risk}_{\text{SVRP}}^S - \text{Risk}_{\text{STDRP}}^S}{\text{Risk}_{\text{SVRP}}^S}, \quad \Delta_T^S := \frac{\text{TT}_{\text{SVRP}}^S - \text{TT}_{\text{STDRP}}^S}{\text{TT}_{\text{SVRP}}^S}, \quad (44b)$$

where the superscript 'S' on the right of Δ represents 'stochastic', while the subscript 'P', 'N', 'R', 'T' represent the indices *Prob*, *Num*, *Risk*, *TT*, respectively. The higher these Δ values, the more benefits truck-and-drone brings.

Fig. 6 shows the values of Δ_P^S , Δ_N^S , Δ_R^S and Δ_T^S in all selected scenarios. One can observe that, employing truck-and-drone tandems can significantly reduce deadline violations in terms of all the four indices, especially when the problem size is large. Moreover, when the sample size decreases, the decision maker has less information on the distribution of travel time, the benefit of truck-and-drone tandems is greater. This phenomenon indicates that introducing drones into delivery systems can better overcome the uncertainty of travel time, and this effect will be amplified when the data size is small.

For robust routing problems, we still test the benefits of truck-and-drone tandems in the four scenarios S_1, S_2, S_3 and S_4 . However, we do not consider the influence of sample size. Instead, we consider the impact of the distributions of testing data on the benefits of truck-and-drone tandems. The experiment procedure is similar to that employed in the previous stochastic case. For each scenario of parameters setting, we generate 200 instances of the samples with a fixed size $K = 100$, drawing from the uniform distribution. For each instance of samples, we estimate the parameters (11)–(12) of the uncertain set and then solve the corresponding RTDRP-D model (13) by Algorithm 2. Denote by its optimal routing solution $\mathbf{x}_{\text{RTDRP}}^*$. Then, we add the constraint $y_i^D = 0, \forall i \in \mathcal{N}$ into the feasible set (1), and find an optimal routing solution $\mathbf{x}_{\text{RVRP}}^*$ to the robust vehicle routing problem with deadlines. Finally, we test the out-of-sample performance of the routing solutions on another 10,000 samples of travel time. Following Section 8.3, we consider four scenarios for the underlying distribution for testing data, i.e., the uniform distribution, the normal distribution, Beta(2,4) distribution and Beta(4,2) distribution.

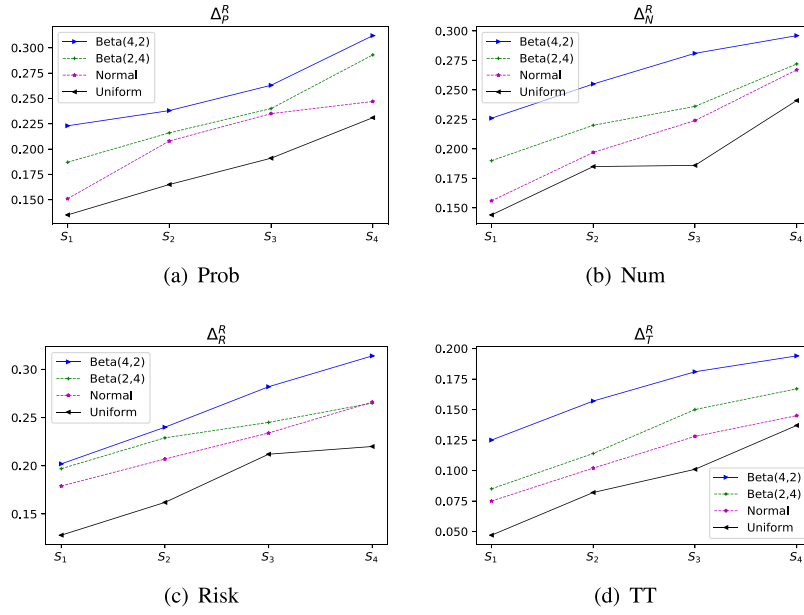


Fig. 7. Benefits of truck-and-drone tandems in robust routing problems.

We repeat the procedure for all the 200 instances, and derive the mean performance index values of $\mathbf{x}_{\text{RTDRP}}^*$ and $\mathbf{x}_{\text{RVRP}}^*$. Then, we define

$$\Delta_P^R := \frac{\text{Prob}_{\text{RVRP}} - \text{Prob}_{\text{RTDRP}}}{\text{Prob}_{\text{RVRP}}}, \quad \Delta_N^R := \frac{\text{Num}_{\text{RVRP}} - \text{Num}_{\text{RTDRP}}}{\text{Num}_{\text{RVRP}}}, \quad (45a)$$

$$\Delta_R^R := \frac{\text{Risk}_{\text{RVRP}} - \text{Risk}_{\text{RTDRP}}}{\text{Risk}_{\text{RVRP}}}, \quad \Delta_T^R := \frac{\text{TT}_{\text{RVRP}} - \text{TT}_{\text{RTDRP}}}{\text{TT}_{\text{RVRP}}}. \quad (45b)$$

These symbols are counterparts of those used in the stochastic case, and we omit their explanations here to avoid duplication.

The testing results for the benefits of truck-and-drone tandems in robust routing problems are reported in Fig. 7. For all the scenarios, deadline violation risk can be mitigated by incorporating truck-and-drone tandems into the delivery system with the uncertainty of travel time. As the size of the delivery network increases, the benefit will be amplified. For instance, the value of Δ_P^R is always increasing when the parameter setting gradually changes from S_1 to S_4 . Moreover, when the distribution of travel time samples is misspecified, the benefit of truck-and-drone tandems will also be amplified. Specifically, if the true distribution of travel time is Beta(4,2), the benefit is the highest in all scenarios. On the contrary, when both the sample data and the real data follow the uniform distribution, that is, when decision-makers can obtain the true distribution of travel time, the benefit of truck-and-drone tandems is the lowest in all scenarios.

In summary, the above experiment results demonstrate that, introducing truck-and-drone tandems in delivery systems can effectively overcome the uncertainty of travel time and reduce the risk of deadline violation, in both the stochastic and robust routing problems.

9. Conclusions

In this study, we address the challenge of mitigating service delay risk in truck-and-drone delivery operations under travel time uncertainty. We handle the uncertainty through a data-driven approach. Based on the samples of travel time, we can construct an empirical distribution to approximate the true distribution, or establish an uncertain set with parameters estimated from the samples. Then, we propose the STDTP-D and RTDTP-D models to help decision-makers find truck-and-drone routing solutions with less delay risk under uncertain environments. For large-scale problems, we introduce BD algorithms for solving the STDTP-D and RTDTP-D models, along with a sample-based routing initialization approach. Our extensive numerical results demonstrate the effectiveness of our proposed models, outperforming benchmarks. Furthermore, we introduce several valid inequalities to enhance the computational efficiency of the BD algorithms.

This research could provide rich managerial insights for logistic operators, especially in the scenario of time-definite delivery. In the process of drone delivery, the uncertainty of travel time and service time cannot be ignored, otherwise it may increase the probability of delivery delay and reduce the service level of logistics enterprises. To overcome the risk of service violation, appropriate modeling of the uncertainty is extremely important for logistic operators. When there is abundant empirical data,

stochastic optimization method can be used to handle the uncertainty; otherwise, when there is limited empirical data, robust optimization method is more suitable.

In practice, the successful implementation of drone delivery largely depends on having well-established landing platforms at customer locations and comprehensive low-altitude airspace management regulations. This requires collaborative efforts from the government, logistics enterprises and the public. As one of the most developed cities in China, Shenzhen is at the forefront of developing low altitude logistics. In 2023, Shenzhen has issued several measures to support high quality development of low-altitude economy. With the support of various government policies, Shenzhen has established an integrated drone-public transportation network, which helps logistic operators overcome operational restrictions of deploying drones and improve service levels by providing rapid package delivery, enhancing the overall safety and efficiency of last mile delivery business.

In addition, public support is also crucial for the effective implementation of drone delivery. Logistics operators need to eliminate the hidden dangers of drone delivery to public safety, as well as the noise pollution caused to residents during the delivery process. This requires logistics operators to develop effective risk management strategies, which can help avoid air collisions and improve the reputation of the enterprise, as it becomes less exposed to uncertainty in reality. Improving the service level not only includes delivering packages to customers within the specified time, but also avoiding noise pollution to other residents by selecting appropriate delivery routes and times.

In conclusion, this study enriches the literature by introducing the decision criterion of service delay risk into truck-and-drone routing problems. However, due to space limitations, there are still many extensions that cannot be discussed in this article and will be left for future research. *First*, the truck-and-drone delivery system can be generalized to a delivery system of multiple trucks and multiple drones. *Second*, besides travel time uncertainty, other significant uncertainties like service time and demand uncertainties need to be considered. *Third*, the proposed model does not consider the affect of energy consumption during the hover and flight of the drone, as it is not the focus of this paper. We refer interested readers to relevant literature (Dukkanci et al., 2021). As a remedy, we extend the model by incorporating the payload and energy consumption constraints in Section 7.2. *Finally*, exploring more general convex risk measures to quantify service lateness risk would add depth to the research. However, addressing these extensions involves increased computational complexity, making them subjects for future investigation.

CRedit authorship contribution statement

Menghua Deng: Software, Methodology. **Yuanbo Li:** Writing – review & editing, Formal analysis. **Jianpeng Ding:** Writing – original draft, Methodology, Conceptualization. **Yanlin Zhou:** Writing – review & editing, Formal analysis. **Lianming Zhang:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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